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Photodetector

Abstract

A sensor includes a first substrate including at least a first pixel. The first pixel includes an avalanche photodiode to convert incident light into electric charge and includes an anode and a cathode. The cathode is in a well region of the first substrate. The first pixel includes an isolation region that isolates the well region from at least a second pixel that is adjacent to the first pixel. The first pixel includes a hole accumulation region between the isolation region and the well region. The hole accumulation region is electrically connected to the anode.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/490,829, filed Sep. 30, 2021, which is a continuation of and claims priority to U.S. patent application Ser. No. 16/905,485, filed Jun. 18, 2020, which is a continuation of and claims priority to U.S. patent application Ser. No. 16/694,195, filed Nov. 25, 2019, now U.S. Pat. No. 10,804,309, which is a continuation of and claims priority to U.S. patent application Ser. No. 16/409,494, filed May 10, 2019, now U.S. Pat. No. 10,580,817, which is a continuation of and claims priority to U.S. patent application Ser. No. 16/060,509, filed Jun. 8, 2018, now U.S. Pat. No. 10,446,601, which is a national stage application under 35 U.S.C. 371 and claims the benefit of PCT Application No. PCT/JP2017/037748 having an international filing date of Oct. 18, 2017, which designated the United States, which PCT application claimed the benefit of Japanese Patent Application No. 2016-204475, filed Oct. 18, 2016; Japanese Patent Application No. 2017-039336, filed Mar. 2, 2017; Japanese Patent Application No. 2017-114380 filed Jun. 9, 2017; and Japanese Patent Application No. 2017-152060, filed Aug. 4, 2017, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present technology relates to a photodetector, for example, a photodetector which is suitable for application to an avalanche photodiode.

BACKGROUND ART

(2) An avalanche photodiode (APD) has a Geiger mode in which it is operated with a bias voltage lower than a breakdown voltage and a linear mode in which it is operated with a slightly high bias voltage in the vicinity of the breakdown voltage. The avalanche photodiode in the Geiger mode is also called single photon avalanche diode (SPAD).

(3) The SPAD is a device capable of detecting a single photon on a pixel basis, by a process in which carriers generated by photoelectric conversion are multiplied in a high-electric-field PN junction region provided on a pixel basis.

(4) In PTL 1, there is proposed a technology in which a layer for forming a high-electric-field region is embedded and is depleted by a bias, for the purpose of enhancing sensitivity of SPAD pixels. In PTL 2, there is proposed a technology for realizing a reduction of crosstalk to neighboring pixels due to light emission in a high-electric-field region, by a structure having an isolation between pixels.

CITATION LIST

Patent Literature

(5) [PTL 1]

(6) JP 2015-41746A

[PTL 2] JP 2013-48278A

SUMMARY OF INVENTION

Technical Problems

(7) In PTL 1, light emission in a high-electric-field region in a pixel at the time of multiplication of carriers may cause a photon to be incident on a neighboring pixel, whereby a signal may be detected in the neighboring pixel in an unintended manner. In addition, a well potential may be varied by a current at the time of multiplication, whereby electric fields in the neighboring pixels may be varied and SPAD characteristics may be varied.

(8) In PTL 2, an isolation region is formed between pixels in order to reduce crosstalk. Physical isolation between the pixels by use of an insulating film is the most effective in reducing the crosstalk. However, it is generally known that a dark current is generated at an interface between the insulating film and silicon. According to PTL 2, therefore, a dark current may be generated. Multiplication of the dark current generated at the interface may worsen DCR (dark count rate).

(9) The present technology has been made in consideration of the above-mentioned circumstances. Accordingly, it is an object of the present technology to enable provision of an APD that is capable of reducing crosstalk and suppressing DCR.

Solution to Problem

(10) According to an aspect of the present technology, a sensor includes a first substrate including at least a first pixel. The first pixel includes an avalanche photodiode to convert incident light into electric charge and includes an anode and a cathode. The cathode is in a well region of the first substrate. The first pixel includes an isolation region that isolates the well region from at least a second pixel that is adjacent to the first pixel. The first pixel includes a hole accumulation region between the isolation region and the well region. The hole accumulation region is electrically connected to the anode.

(11) According to a second aspect of the present technology, a sensor includes substrate including at least one pixel. The at least one pixel includes an avalanche photodiode to convert incident light into electric charge and includes a cathode of a first conductivity type and an anode of a second conductivity type. The cathode is in a well region of the first conductivity type, and the anode is at a perimeter of the well region. The at least one pixel includes a first material of the second conductivity type around the well region. The first material is electrically connected to the anode. The at least one pixel includes an insulating material around the first material and the anode.

(12) According to a third aspect of the present technology, a sensor includes a first substrate including at least a first pixel in a pixel region. The first pixel includes an avalanche photodiode to convert incident light to electric charge and includes a first region, a second region, and a third region. The first and third regions are a first conductivity type and are stacked. The second region is a second conductivity type. The first pixel includes an isolation region that isolates the third region from at least a second pixel, wherein the isolation region penetrates through the first substrate. The first pixel includes a charge accumulation region of the second conductivity type between the third region and the isolation region. The charge accumulation region is electrically connected to the second region.

(13) In at least one additional aspect of the present technology, a high-electric-field region, an isolation region for isolation from neighboring pixels, and an electron accumulation region that accumulates electrons on a side wall of the isolation region are provided, and the electron accumulation region is electrically connected with an anode.

Advantageous Effects of Invention

(14) According to an aspect of the present technology, an APD capable of reducing crosstalk and suppressing DCR can be provided.

(15) Note that the effect described here is not necessarily restrictive, and any of the effects described herein may be attained.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 depicts a configuration of a first embodiment of an APD to which the present technology has been applied.
- (2) FIG. 2 depicts another configuration of the APD.
- (3) FIG. 3 is a sectional view of the APD.
- (4) FIG. 4 is a plan view of the APD.
- (5) FIG. 5 is a plan view of the APD.
- (6) FIG. 6 is a sectional view of another configuration of the APD.
- (7) FIG. 7 is a sectional view of another configuration of the APD.
- (8) FIG. 8 is a sectional view of another configuration of the APD.
- (9) FIG. 9 is a sectional view of another configuration of the APD.
- (10) FIG. 10 is a sectional view of another configuration of the APD.
- (11) FIG. 11 is a sectional view of another configuration of the APD.
- (12) FIG. 12 is a sectional view of another configuration of the APD.
- (13) FIG. 13 is a sectional view of another configuration of the APD.
- (14) FIG. 14 is a sectional view of another configuration of the APD.
- (15) FIG. 15 is a sectional view of another configuration of the APD.
- (16) FIG. 16 is a plan view of another configuration of the APD.
- (17) FIG. 17 is a plan view of another configuration of the APD.
- (18) FIG. 18 includes a sectional view of another configuration of the APD.
- (19) FIG. 19 is a sectional view of another configuration of the APD.
- (20) FIG. 20 is a plan view of another configuration of the APD.
- (21) FIG. 21 includes a sectional view of another configuration of the APD.
- (22) FIG. 22 is a plan view of another configuration of the APD.
- (23) FIG. 23 is a sectional view of another configuration of the APD.
- (24) FIG. 24 is a sectional view of another configuration of the APD.
- (25) FIG. 25 is a sectional view of another configuration of the APD.
- (26) FIG. 26 is a figure for explaining movement of an electron.
- (27) FIG. 27 is a figure for explaining movement of an electron.
- (28) FIG. 28 includes a figure for explaining the position of a block layer.
- (29) FIG. 29 is a sectional view of another configuration of the APD.
- (30) FIG. 30 is a sectional view of another configuration of the APD.
- (31) FIG. 31 is a sectional view of another configuration of the APD.
- (32) FIG. 32 is a plan view of another configuration of the APD.
- (33) FIG. 33 is a figure for explaining another configuration of an n-type semiconductor region.
- (34) FIG. 34 is a figure for explaining movement of an electron.
- (35) FIG. 35 is a figure for explaining a pixel region, a peripheral region, and a pad region.
- (36) FIG. 36 is a sectional view of an APD.
- (37) FIG. 37 is a sectional view of a configuration of an isolation region.
- (38) FIG. 38 is a sectional view of another configuration of an isolation region.
- (39) FIG. 39 is a sectional view of another configuration of an isolation region.
- (40) FIG. 40 is a sectional view of another configuration of an isolation region.
- (41) FIG. 41 is a sectional view of another configuration of an isolation region.
- (42) FIG. 42 is a sectional view of another configuration of an isolation region.

- (43) FIG. **43** is a sectional view of another configuration of an isolation region.
- (44) FIG. **44** is a sectional view of another configuration of an isolation region.
- (45) FIG. **45** is a sectional view of another configuration of an isolation region.
- (46) FIG. **46** is a sectional view of another configuration of an isolation region.
- (47) FIG. **47** is a sectional view of another configuration of an isolation region.
- (48) FIG. **48** is a sectional view of another configuration of an isolation region.
- (49) FIG. **49** is a sectional view of another configuration of an isolation region.
- (50) FIG. **50** is a sectional view of another configuration of an isolation region.
- (51) FIG. **51** is a sectional view of a configuration of a peripheral region.
- (52) FIG. **52** is a plan view for explaining a shape of a trench in the configuration of the peripheral region.
- (53) FIG. **53** is a plan view for explaining shapes of a trench in the configuration of the peripheral region.
- (54) FIG. **54** is a sectional view of another configuration of a peripheral region.
- (55) FIG. **55** is a plan view for explaining a shape of a trench in the configuration of the peripheral region.
- (56) FIG. **56** is a sectional view of another configuration of a peripheral region.
- (57) FIG. **57** is a sectional view of another configuration of a peripheral region.
- (58) FIG. **58** is a sectional view of another configuration of a peripheral region.
- (59) FIG. **59** is a sectional view of another configuration of a peripheral region.
- (60) FIG. **60** is a sectional view of another configuration of a peripheral region.
- (61) FIG. **61** is a figure for explaining a shape of a trench.
- (62) FIG. **62** is a figure for explaining a shape of a trench.
- (63) FIG. **63** is a sectional view of another configuration of a peripheral region.
- (64) FIG. **64** is a figure illustrating a configuration of an imaging apparatus.
- (65) FIG. **65** is a figure for explaining a TOF.
- (66) FIG. **66** is a block diagram depicting an example of general configuration of a vehicle control system.
- (67) FIG. **67** illustrates an example of layout positions of exterior information detection units and imaging sections.

DESCRIPTION OF EMBODIMENTS

- (68) Modes for carrying out the present technology (hereinafter referred to as embodiments) will be described below.
- (69) The present technology as described below is applicable to a photodetector. In addition, since the present technology is applicable particularly to an avalanche photodiode (APD) as a photodetector, description will be made here taking an APD as an example. An APD has a Geiger mode in which it is operated with a bias voltage lower than its breakdown voltage and a linear mode in which it is operated at a slightly high bias voltage in the vicinity of the breakdown voltage. The APD in the Geiger mode is also called single photon avalanche diode (SPAD).
- (70) The SPAD is a device capable of detecting a single photon on a pixel basis, by a process in which carriers generated by photoelectric conversion are multiplied in a high-electric-field PN junction region provided on a pixel basis. When the present technology is applied to an SPAD among APDs, a higher effect can be obtained.
- (71) <Front Side Illumination Type Pixel>
- (72) FIG. **1** depicts a configuration of a front side illumination type APD. A pixel **10**, when referred to from the lower side in the figure, has a configuration in which a wiring layer **22** is stacked on an APD **21**, and an on-chip lens **23** is stacked on the wiring layer **22**. Detailed configuration of the APD **21** will be described later.
- (73) Light is incident from the on-chip lens **23** side (in FIG. **1**, the upper side), and is incident on the APD **21** through the wiring layer **22**. In the case of the front side illumination type APD **10**, a

circuit is disposed in a region outside a pixel area, and the circuit is disposed in a same substrate.

(74) <Backside Illumination Type APD>

(75) FIG. 2 depicts a configuration of a backside illumination type pixel. A pixel **30**, when referred to from the lower side in the figure, has a configuration in which an APD **21** is stacked on an on-chip lens **23**, a sensor substrate **41** is stacked on the APD **21**, and a circuit substrate **42** is stacked on the sensor substrate **41**.

(76) Light is incident from the on-chip lens **23** side (in FIG. 2, the lower side), and is incident on the APD **21**. In the case of the backside illumination type pixel **30**, a configuration in which the circuit is stacked as the circuit substrate **42** as depicted in FIG. 2 may be adopted, or a configuration in which the circuit is disposed in a region outside the pixel area and is disposed in the same substrate may be adopted.

(77) The present technology is applicable to both the front side illumination type pixel **10** as depicted in FIG. 1 and the backside illumination type pixel **30** as depicted in FIG. 2. The following description will be made taking the backside illumination type pixel **30** as an example. In the following description, only the APD **21** is illustrated, and the on-chip lens **23** is omitted from illustration.

Configuration of APD in First Embodiment

(78) FIG. 3 depicts a sectional configuration in a first embodiment of the APD **21**. As depicted in FIG. 3, an APD **21a** includes an n-type semiconductor region **101** of which the conductivity type is n type (first conductivity type), and a p-type semiconductor region **102** of which the conductivity type is p type (second conductivity type) and which is formed under the n-type semiconductor region **101**. The n-type semiconductor region **101** and the p-type semiconductor region **102** are formed in a well layer **103**.

(79) The well layer **103** may be a semiconductor region of n-type conductivity, or may be a semiconductor region of p-type conductivity. In addition, the well layer **103** is preferably an n-type or p-type semiconductor region of a low concentration of not more than 1E14 order, for example, whereby the well layer **103** can be easily depleted, and enhanced PDE can be realized.

(80) The n-type semiconductor region **101** is a semiconductor region of n-type conductivity which includes, for example, Si (silicon) and has a high impurity concentration. The p-type semiconductor region **102** is a semiconductor region of p-type conductivity which has a high impurity concentration. The p-type semiconductor region **102** constitutes a pn junction at its interface with the n-type semiconductor region **101**. The p-type semiconductor region **102** has a multiplication region in which carriers generated by incidence of light to be detected are put to avalanche multiplication. The p-type semiconductor region **102** is preferably in a depleted state, whereby enhanced PDE can be realized.

(81) The n-type semiconductor region **101** functions as a cathode, and is connected to a circuit through a contact **104**. An anode **105** opposed to the cathode is formed in the same layer as the n-type semiconductor region **101**, between the n-type semiconductor region **101** and an isolation region **108**. The anode **105** is connected to a circuit through a contact **106**.

(82) The isolation region **108** for isolating the APDs **21a** from each other is formed, and a hole accumulation region **107a** is formed between the isolation region **108** and the well layer **103**. The hole accumulation region **107a** is formed on the lower side of the anode **105**, in the state of being electrically connected with the anode **105**. In addition, the hole accumulation region **107a** is formed between the well layer **103** and the isolation region **108**. Further, the hole accumulation region **107a** is also formed under the well layer **103** (on the back side of the APD **21a**).

(83) The hole accumulation region **107a** is formed at a part where different materials make contact. In the example depicted in FIG. 3, the isolation region **108** includes, for example, a silicon oxide film, which is a different material from the well layer **103**. Therefore, the hole accumulation region **107a** is formed for suppressing a dark current generated at the interface.

(84) In addition, in the case where the APD **21a** is applied to the backside illumination type, for

example as depicted in FIG. 2, the on-chip lens **23** is stacked on the lower side (on the side opposite to the side where the n-type semiconductor region **101** is formed) of the well layer **103**, and, in this case, the hole accumulation region **107a** is formed also at the interface with the well layer **103** on the side where the on-chip lens **23** is formed.

(85) Besides, in the case where the APD **21a** is applied to the front side illumination type, for example as depicted in FIG. 1, the silicon substrate is present on the lower side (on the side opposite to the side where the n-type semiconductor region **101** is formed) of the well layer **103**, and, therefore, a configuration may be adopted in which the hole accumulation region **107a** is not formed. Naturally, even in the case where the APD **21a** is applied to the front side illumination type, the hole accumulation region **107a** may be formed on the lower side of the well layer **103**.

(86) In other words, the hole accumulation region **107a** may be formed on other surfaces than the upper surface (the surface on which the n-type semiconductor region **101** is formed) of the well layer **103**. Alternatively, the hole accumulation region **107a** may be formed on other surfaces than the upper and lower surfaces of the well layer **103**.

(87) The hole accumulation region **107a** may be formed as a p-type semiconductor region.

(88) The isolation region **108** is formed between the APDs **21a**, to isolate each of the APDs **21a**. Specifically, the isolation region **108** is formed in such a manner that multiplication regions are formed in one-to-one correspondence with the APDs **21a**. As depicted in FIG. 4, the isolation region **108** is formed in a two-dimensional grid pattern such as to perfectly surround the periphery of each multiplication region (APD **21a**). As illustrated in FIG. 3, the isolation region **108** is formed to penetrate from the upper surface side to the lower surface side of the well layer **103** in the stacking direction. Note that other configuration than the configuration in which the isolation region **108** entirely penetrates from the upper surface side to the lower surface side of the well layer **103**, for example, a configuration in which the isolation region **108** penetrates only part of the well layer **103** and is inserted into an intermediate portion of the substrate may be adopted.

(89) FIG. 4 is a plan view of the APD **21a**, as viewed from the upper surface side (the upward side in FIG. 3, namely, the n-type semiconductor region **101** side). In FIG. 4, an example is depicted in which four (2×2) APDs **21a-1** to **21a-4** are arranged.

(90) As aforementioned, the APDs **21a** are isolated from one another by the isolation region **108** formed in a grid pattern. The anodes **105** are formed inside the isolation region **108**. The well layer **103** is formed between the anode **105** and the n-type semiconductor region **101**. The n-type semiconductor region **101** is formed in a central area of the APD **21a**.

(91) Although the hole accumulation region **107a** is not seen when viewed from the upper surface side, the hole accumulation region **107a** is formed inside the isolation region **108**. In other words, the hole accumulation region **107a** is formed in a region which is substantially the same as the region of the anode **105**.

(92) While an example in which the n-type semiconductor region **101** is formed in a tetragonal shape is illustrated in FIG. 4, the n-type semiconductor region **101** may be formed in a circular shape, as illustrated in FIG. 5.

(93) In the case where the n-type semiconductor region **101** is formed in a tetragonal shape as depicted in FIG. 4, the area of the multiplication region (n-type semiconductor region **101**) can be secured to be large, so that detection efficiency called PDE (Photon Detection Efficiency) can be enhanced.

(94) In the case where the n-type semiconductor region **101** is formed in a circular shape as depicted in FIG. 5, electric field concentration at edge portions can be suppressed, so that unintentional edge breakdown can be reduced.

(95) Description in the following will be made taking as an example a case in which the n-type semiconductor region **101** is formed in a tetragonal shape as depicted in FIG. 4, but the description may be applied also to a case in which the n-type semiconductor region **101** is formed in a circular shape as depicted in FIG. 5.

(96) With the hole accumulation region **107a** thus formed at the interface, electrons generated at the interface can be trapped, and DCR (dark count rate) can be suppressed. In addition, while the hole accumulation region **107a** is described and description is made here taking as an example a case in which holes are accumulated and electrons are trapped, a configuration may be adopted in which electrons are accumulated and holes are trapped, as in an APD **21j** (FIG. **23**) that will be described as a tenth embodiment. In the case where holes are trapped, also, it is possible to suppress DCR.

(97) In addition, with the isolation region **108** and the hole accumulation region **107a** provided, it is possible to reduce electrical crosstalk and optical crosstalk. Besides, with the hole accumulation region **107a** provided on side surfaces, lateral electric fields are generated, carriers can be collected into high-electric-field regions more easily, and PDE can be enhanced.

(98) The hole accumulation region **107a** can be formed by ion injection, solid-phase diffusion, induction by a fixed charge film, or the like.

Configuration of APD in Second Embodiment

(99) FIG. **6** depicts a sectional configuration in a second embodiment of the APD **21**. Of an APD **21b** depicted in FIG. **6**, the parts similar to those of the APD **21a** depicted in FIG. **3** are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(100) The APD **21b** illustrated in FIG. **6** differs from the APD **21a** depicted in FIG. **3** in the size of the region where the hole accumulation region **107** is formed. A hole accumulation region **107b** of the APD **21b** depicted in FIG. **6** is formed only at part of the portion where the isolation region **108** is formed.

(101) As depicted in FIG. **6**, the hole accumulation region **107b** is formed in a region of at least part of a side wall of the isolation region **108**. In addition, the hole accumulation region **107b** thus formed is formed in the state of being electrically connected with the anode **105**.

(102) When the APD **21b** depicted in FIG. **6** is viewed from the upper surface side, it is seen as depicted in FIG. **4** or FIG. **5**. Besides, also in the APD **21b**, like in the APD **21a** (FIG. **3**), the hole accumulation region **107b** is formed under the anode **105**. Like the anode **105** which is formed inside the isolation region **108** in such a manner as to continuously surround the well layer **103**, the hole accumulation region **107b** is also formed inside the isolation region **108** in such a manner as to continuously surround the well layer **103**.

(103) Also in the case where the hole accumulation region **107** is formed in this way, the formation of the hole accumulation region **107b** in the vicinity of a high-electric-field region ensures that generation of a dark current from a portion near the high-electric-field region can be restrained, and DCR can be suppressed.

(104) Also in the APD **21b**, like in the APD **21a** depicted in FIG. **3**, there can be expected such effects as a reduction of crosstalk, an effect by which carriers can be easily collected into the high-electric-field region, and an effect by which PDE can be enhanced.

(105) In addition, in the case where the hole accumulation region **107b** is formed by ion injection, it is possible to reduce the number of ion injection steps required. For instance, in a case where the hole accumulation region **107a** is formed ranging to a depth portion of a silicon substrate, as in the APD **21a** depicted in FIG. **3**, ion injection should be carried out multiple times, for example, n times. On the other hand, in a case where the hole accumulation region **107b** is formed in part of a silicon substrate, as in the APD **21b** depicted in FIG. **6**, it is sufficient to carry out ion injection a number of times less than n times; thus, the number of ion injection steps required can be reduced, and process cost can be lowered.

(106) According to the APD **21b** in the second embodiment, DCR and process cost can be well balanced.

Configuration of APD in Third Embodiment

(107) FIG. **7** depicts a sectional configuration in a third embodiment of the APD **21**. Of an APD **21c** illustrated in FIG. **7**, the parts similar to those of the APD **21a** depicted in FIG. **3** are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(108) The APD **21c** depicted in FIG. 7 differs from the APD **21a** depicted in FIG. 3 in that a fixed charge film that generates a hole accumulation region **107** is formed. A fixed charge film **151** is formed at side surfaces inside the isolation region **108**, unlike the hole accumulation region **107a** of the APD **21a** depicted in FIG. 3. In addition, in the example depicted in FIG. 7, the fixed charge film **151** is formed also on the back surface side of the APD **21c**.

(109) Note that, like the hole accumulation region **107a** of the APD **21a** depicted in FIG. 3, the fixed charge film **151** also may not be formed on the back surface side of the APD **21c**. In other words, the fixed charge film **151** may be formed only at the side surfaces inside the isolation region **108**.

(110) In addition, like in FIG. 6 (second embodiment), the fixed charge film **151** may be formed at part of the isolation region **108**.

(111) The fixed charge film **151** is a negative fixed charge film. On the inside of the fixed charge film **151**, a hole accumulation region **107c** is formed due to induction of holes by the fixed charge film **151**. Since the hole accumulation region **107c** is thus formed, there can be obtained such effects as suppression of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced, like in the first and second embodiments.

(112) In addition, it becomes unnecessary to form the hole accumulation region **107c** by carrying out ion injection multiple times in a range to a depth portion of silicon; thus, the number of ion injection steps required can be reduced.

(113) As illustrated in FIG. 8, the first embodiment (FIG. 3) and the third embodiment (FIG. 7) may be combined with each other. Specifically, a configuration may be adopted in which the fixed charge film **151** is formed at side surfaces inside the isolation region **108**, and a hole accumulation region **107c'** (the dash is affixed for distinction from the hole accumulation region **107c** formed by the fixed charge film **151**) is formed at side surfaces of the fixed charge film **151**.

(114) Besides, a configuration may be adopted in which the fixed charge film **151** and the hole accumulation region **107c'** are formed also on the back surface side of the APD **21c** or in which they are not formed on the back surface side of the APD **21c**.

(115) When the APD **21c** depicted in FIG. 7 or FIG. 8 is viewed from the upper surface side, it is seen as depicted in FIG. 4 or FIG. 5. In addition, like in the APD **21a** (FIG. 3), in the APD **21c** also, the fixed charge film **151** (the hole accumulation region **107c**, the hole accumulation region **107c'**) is formed under the anode **105**. Besides, like the anode **105** formed at side walls of the isolation region **108** such as to continuously surround the well layer **103**, the fixed charge film **151** (the hole accumulation region **107c**, the hole accumulation region **107c'**) is also formed at side walls of the isolation region **108** such as to continuously surround the well layer **103**.

(116) According to the configuration illustrated in FIG. 8, the hole accumulation region **107c'** due to the fixed charge film **151** and the hole accumulation region **107c** are formed. Therefore, a dark current can be suppressed more assuredly, and a further suppression of DCR can be realized.

Configuration of APD in Fourth Embodiment

(117) FIG. 9 depicts a sectional configuration in a fourth embodiment of the APD **21**. Of an APD **21d** depicted in FIG. 9, the parts similar to those of the APD **21a** depicted in FIG. 3 are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(118) The APD **21d** illustrated in FIG. 9 differs from the APD **21a** depicted in FIG. 3 in that a metallic film **171** is formed in the isolation region **108**.

(119) FIG. 10 is a plan view of the APD **21d** as viewed from the upper surface side (the upward side in FIG. 3, or the n-type semiconductor region **101** side), depicting an example in which four (2×2) APDs **21d-1** to **21d-4** are arranged.

(120) Like in the first to third embodiments, the isolation region **108** is formed between a plurality of APDs **21d**, to isolate each APD **21d**. As depicted in FIG. 10, the isolation region **108** is formed in a two-dimensional grid pattern such as to perfectly surround the periphery of each APD **21d**.

- (121) Further, the metallic film **171** is formed at a central portion of the isolation region **108**. Like the isolation region **108**, the metallic film **171** is formed between the APDs **21d**, is formed at a position for isolating each APD **21d**, and is formed in a two-dimensional grid pattern such as to perfectly surround the periphery of each APD **21d**.
- (122) As depicted in FIG. **9**, the isolation region **108** and the metallic film **171** are formed to penetrate from the upper surface side to the lower surface side of the well layer **103** in the stacking direction.
- (123) While an example in which the metallic film **171** is formed in the APD **21a** described referring to FIG. **3**, for example, in the first embodiment has been taken in describing the APD **21d** referring to FIG. **9** and FIG. **10**, there may be adopted a configuration in which the metallic film **171** is formed in the APD **21b** described referring to FIG. **6** in the second embodiment.
- (124) In addition, a configuration may be adopted in which the metallic film **171** is formed in the APD **21c** (APD **21c'**) described referring to FIG. **7** or **8** in the third embodiment.
- (125) With the isolation region **108** and the metallic film **171** thus formed between the APDs **21d**, influences exerted from neighboring APDs **21d** can be reduced. For example, crosstalk due to light emission inside the APD **21d** can be further reduced.
- (126) A configuration may be adopted in which the metallic film **171'** is formed inside the isolation region **108**, and a hole accumulation region **107d** is formed at side walls of the isolation region **108**, as depicted in FIG. **9**. Alternatively, a configuration may be adopted in which a metallic film **171** is formed inside the isolation region **108**, but the hole accumulation region **107d** is not formed at the side surfaces, as depicted in FIG. **11**.
- (127) As illustrated in FIG. **11**, the metallic film **171'** is formed on a side surface on one side of the isolation region **108**, and a voltage is impressed on the metallic film **171'**. With the voltage impressed on the metallic film **171'**, a hole accumulation region **107d'** (not depicted in FIG. **11**) is formed in the vicinity of the metallic film **171'**.
- (128) Also in the case of this configuration in which a voltage is impressed on the metallic film **171'** to form the hole accumulation region **107d'** (not depicted) in the vicinity of the metallic film **171'**, a configuration is realized in which electrons generated at the interface between the well layer **103** and the isolation region **108** can be trapped, so that a dark current generated at the interface can be reduced.
- (129) In the APD **21d** depicted in FIG. **9**, also, it is possible to impress a voltage on the metallic film **171** to thereby form the hole accumulation region **107d'** in the vicinity of the metallic film **171**. Where the hole accumulation region **107d** formed by ion injection and the hole accumulation region **107d'** formed by the metallic film **171** are combined with each other, a further reduction of DCR can be achieved.
- (130) In addition, though not illustrated, where a fixed charge film **151** is formed like in the third embodiment and the hole accumulation region **107c** formed by the fixed charge film **151** and the metallic film **171** (the hole accumulation region **107d'** formed thereby) are combined with each other, a further reduction in DCR can be achieved.

Configuration of APD in Fifth Embodiment

- (131) FIG. **12** depicts a sectional configuration in a fifth embodiment of the APD **21**. Of an APD **21e** illustrated in FIG. **12**, the parts similar to those of the APD **21a** depicted in FIG. **3** are denoted by the same reference symbols as used above, and descriptions of them are omitted.
- (132) The APD **21e** depicted in FIG. **12** differs from the APD **21a** depicted in FIG. **3** in that a well layer **103e** is configured to be larger in thickness (depth).
- (133) A depth region of the well layer **103e** (silicon substrate) is depleted from the back surface to a high-electric-field region by application of a bias. As the silicon film thickness is larger, quantum efficiency at long wavelengths such as near infrared light is enhanced. However, as the place where carriers are generated by photoelectric conversion becomes far from the high-electric-field region, PDE is lowered. Therefore, the silicon film thickness should be set while taking a balance of

characteristics.

(134) Even in the case where the silicon film thickness is enlarged, the hole accumulation region **107a** can be formed in such a manner as to surround the well layer **103**, like in the first embodiment (FIG. 3).

(135) As has been described in the second embodiment, however, for forming the hole accumulation region **107a** ranging to a depth portion of the silicon substrate it may be necessary to carry out ion injection multiple times, leading to an increase in the number of steps. In addition, an acceleration energy for ion injection is limited on a process basis, and, therefore, it is difficult to form the hole accumulation region **107a** ranging to the depth portion of the silicon substrate having a large silicon film thickness.

(136) In view of this, in the case where the silicon substrate is thick, a configuration in which the hole accumulation region **107a** is formed in the vicinity of a part where the n-type semiconductor region **101** is formed in the well layer **103** and in which the hole accumulation region **107a** is not formed in a depth portion of the silicon substrate, may be adopted, like in the second embodiment (FIG. 6).

(137) However, in a depth portion of the silicon substrate, it may be impossible to suppress DCR without providing a mechanism for suppressing DCR. Therefore, a mechanism for suppressing DCR is provided also in the depth portion of the silicon substrate. As an example of such a mechanism, a fixed charge film **151** is formed, like in the third embodiment (FIG. 7).

(138) Further, for reducing influences of neighboring APDs **21e**, a metallic film **171** is formed in the isolation region **108**, like in the fourth embodiment (FIG. 9).

(139) An APD **21e** illustrated in FIG. 12 depicts an example in which the silicon film thickness is large and the second to fourth embodiments are combined together. Specifically, the APD **21e** depicted in FIG. 12 has a configuration in which a metallic film **171e** is formed in the isolation region **108**, and a fixed charge film **151e** is formed at a side surface on the well layer **103e** side. In addition, a hole accumulation region **107e** is formed at part of side surfaces on the well layer **103e** side of the fixed charge film **151e**.

(140) Besides, the fixed charge film **151e** is formed also on the back surface side of the APD **21e**. Note that a configuration may be adopted in which the fixed charge film **151e** is not formed on the back surface side of the APD **21e**.

(141) Thus, in the case where the silicon film thickness of the APD **21e** is large, a structure in which the hole accumulation region **107e** and the fixed charge film **151e** are formed on the high-electric-field region side where the n-type semiconductor region **101** is formed gives a configuration in which electrons generated at the interface between the well layer **103e** and the isolation region **108** can be trapped, so that a dark current generated at the interface can be reduced.

(142) In addition, in the case where the silicon film thickness of the APD **21e** is large, a structure in which the fixed charge film **151e** is formed in a depth portion (on the back surface side) of the APD **21e** gives a configuration in which the hole accumulation region **107** (not depicted) due to the fixed charge film **151e** is formed, and electrons generated at the interface between the well layer **103e** and the isolation region **108** can be trapped, so that a dark current generated at the interface can be reduced.

(143) Further, a voltage may be impressed on the metallic film **171e** such that a hole accumulation region **107e'** is formed owing to the metallic film **171e**. In this case, electrons can be trapped also in the hole accumulation region **107e'** owing to the metallic film **171e**, so that the dark current can be further reduced.

(144) With the configuration as depicted in FIG. 12, the dark current can be reduced and DCR can be suppressed, in the whole part of the APD **21e** having a large silicon film thickness. Besides, like in the first to fourth embodiments, there can be obtained such effects as a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(145) When the APD **21e** illustrated in FIG. **12** is viewed from the upper surface side (the side where the n-type semiconductor region **101** is formed), the APD **21e** appears to have the metallic film **171e** formed at a substantially central portion of the isolation region **108**, as depicted in FIG. **10**.

(146) As illustrated in FIG. **13**, a configuration may be adopted which is obtained by eliminating the hole accumulation region **107e** from the APD **21e** depicted in FIG. **12**. In an APD **21e'** depicted in FIG. **13**, the metallic film **171e** is formed in the isolation region **108**, and the fixed charge film **151e** is formed on the well layer **103e** side.

(147) In such a configuration, also, a hole accumulation region **107e''** is formed owing to the fixed charge film **151e**, so that there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced. Naturally, the hole accumulation region **107e** formed by ion injection may be combined with this, whereby a further reduction of DCR can be achieved.

(148) In addition, with a voltage impressed on the metallic film **171e**, the hole accumulation region **107e''** is formed, so that there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(149) As illustrated in FIG. **14**, a configuration may be adopted which is obtained by eliminating the fixed charge film **151e** from the APD **21e** depicted in FIG. **12**. In an APD **21e''** depicted in FIG. **14**, the metallic film **171e** is formed in the isolation region **108**, and the hole accumulation region **107e** is formed at side walls of the well layer **103e**.

(150) In the case of such a configuration, also, since the hole accumulation region **107e** is formed, there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(151) In addition, application of a voltage to the metallic layer **171e** forms a hole accumulation region (the hole accumulation region formed by applying a voltage to the metallic layer **171e** is not depicted in FIG. **14**), so that there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions, and an effect by which PDE can be enhanced.

Configuration of APD in Sixth Embodiment

(152) FIG. **15** depicts a sectional configuration in a sixth embodiment of the APD **21**. Of an APD **21f** illustrated in FIG. **15**, the parts similar to those of the APD **21a** depicted in FIG. **3** are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(153) The APD **21f** depicted in FIG. **15** differs from the APD **21a** depicted in FIG. **3** in the shape of the n-type semiconductor region **101** and in the position where the n-type semiconductor region **101** is formed. Of the n-type semiconductor region **101f** of the APD **21f** depicted in FIG. **15**, other portions than a portion connected with the contact **104** are embedded in the well layer **103**. In the example illustrated in FIG. **15**, the n-type semiconductor region **101** is formed such as to have a projected portion at a substantially central portion.

(154) In the APD **21f** depicted in FIG. **15**, a high concentration diffusion layer (in this case, the n-type semiconductor region **101** functioning as a cathode) that takes electrical contact is formed separate from a high concentration diffusion layer (in this case, an anode **105**) of opposite conductivity type that takes a different potential.

(155) Referring to FIG. **15**, the n-type semiconductor region **101f** is formed with a projected portion (hereinafter referred to as projected portion **101f**), and the n-type semiconductor region **101f** as a whole inclusive of the projected portion **101f** is formed at a position separate from the anode **105**.

(156) When the APD **21f** is viewed from the upper surface side (the n-type semiconductor region

101f side), it is seen as depicted in FIG. **16**. In FIG. **16** is depicted an example in which four (2×2) APDs **21f-1** to **21f-4** are arranged. When one APD **21f** is viewed from above, the projected portion **101f** of the n-type semiconductor region **101f** to be connected with the contact **104** is disposed in a central area thereof, and the anode **105** is formed at a position separate from the projected portion **101f** such as to surround the projected portion **101f**.

(157) Between the n-type semiconductor region **101f** and the anode **105**, the n-type semiconductor region **101** may be formed in an exposed state, or may be configured to be covered with a semiconductor region **191** of an opposite conductivity type, as depicted in FIG. **15**. The semiconductor region **191** of the opposite conductivity type may be a p-type semiconductor region **191**, for example. Besides, the semiconductor region **191** of the opposite conductivity type may also be a layer formed by ion injection, like a hole accumulation region **107f**, for example.

(158) In this way, of a region constituting the high-electric-field region, other portions than a portion connected with a wiring for the cathode are formed in the silicon substrate, and the semiconductor region **191** (hole accumulation region) is formed on a surface of the silicon substrate, in a region between the portion (projected portion **101f**) connected with the wiring for the cathode and the anode **105**.

(159) Thus, the area of a high-concentration impurity region connected with the contact **104** (in this case, the n-type semiconductor region **101f** connected with the contact **104**) is reduced, whereby DCR can be suppressed. Besides, in this case, the shape of the n-type semiconductor region **101f** forming the cathode is set to be the shape having the projected portion as depicted in FIG. **15**, whereby PDE can be maintained, since the area of the high-electric-field region itself is not reduced.

(160) In addition, like in the first to fifth embodiments, there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(161) Besides, while an example in which the sixth embodiment is applied to the first embodiment (FIG. **3**) has been described here, the sixth embodiment, specially, the configuration in which the n-type semiconductor region **101** having a projected portion is formed and the substrate surface is covered with the semiconductor region **191** of the opposite conductivity type, may be applied to the second to fifth embodiments.

Configuration of APD in Seventh Embodiment

(162) FIG. **17** depicts a plan-view configuration in a seventh embodiment of the APD **21**. Of an APD **21g** illustrated in FIG. **17**, the parts similar to those of the APD **21a** depicted in FIG. **4** are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(163) The APD **21g** depicted in FIG. **17** differs from the APD **21a** depicted in FIG. **4** in that the area of the anode **105g** is reduced.

(164) FIG. **17** is a plan view of the APD **21g** in the case where the APD **21g** is viewed from the upper surface side (the n-type semiconductor region **101** side), and depicts an example in which four (2×2) APDs **21g-1** to **21g-4** are arranged. Referring to the APD **21g-1**, for example, the anode **105g** of the APD **21g-1** is formed only at four corners adjacent to the isolation region **108**.

(165) Of the APD **21g** depicted in FIG. **17**, a sectional view taken along line A-A' is depicted in FIG. **18A**, while a sectional view taken along line B-B' is depicted in FIG. **18B**. FIG. **18A** depicts a section of a portion where the anode **105g** is not formed, whereas FIG. **18B** depicts a section of a portion where the anode **105g** is formed.

(166) As depicted in FIG. **18A**, there is a portion where the anode **105g** is not formed, and as depicted in FIG. **18B**, there is a portion where the anode **105** is formed. Thus, the anodes **105g** are formed at parts of the inner periphery of the isolation region **108**. As a result, the area of a high-concentration diffusion layer taking electrical contact (in this case, the area of the anode **105g** connected with the contact **106**) can be reduced, and this high-concentration diffusion layer can be

separated from a high-concentration diffusion layer of an opposite conductivity type that takes another potential (in this case, the n-type semiconductor region **101** constituting the cathode connected with the contact **104**).

(167) With such a configuration, damage defects can be reduced, electric field relaxation can be realized, and DCR can be suppressed.

(168) In addition, since the hole accumulation region **107** is formed, like in the first to sixth embodiments, there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(169) Note that while an example in which the anode **105g** of the APD **21g** is formed only at each of the four corners inside the isolation region **108** has been depicted in the description here, the anode **105g** may be formed in other region than the four corners, for example, only at part of the inner periphery of the isolation region **108**, such as only at an edge (side) inside the isolation region **108**.

(170) While an example in which the seventh embodiment is applied to the first embodiment (FIG. 3) has been depicted here, the seventh embodiment, specifically, the configuration in which the anode **105** is formed at only part of the inner periphery of the isolation region **108**, may be applied to the second to sixth embodiments.

Configuration of APD in Eighth Embodiment

(171) FIG. **19** depicts a sectional configuration in an eighth embodiment of the APD **21**. Of an APD **21h** illustrated in FIG. **19**, the parts similar to those of the APD **21a** depicted in FIG. 3 are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(172) The APD **21h** depicted in FIG. **19** differs from the APD **21a** depicted in FIG. 3 in the shape of the n-type semiconductor region **101h**. Of an n-type semiconductor region **101h** of the APD **21h** depicted in FIG. **19**, other portions than a portion connected with a contact **104h** are embedded in the well layer **103**. In the example illustrated in FIG. **19**, the n-type semiconductor region **101** is formed to have a projected portion.

(173) Note that while the projected portions are formed on the left and right sides of the n-type semiconductor region **101h** in FIG. **19** since FIG. **19** is a sectional view of the APD **21a**, the projected portion is formed continuously to form a ring in plan view, as depicted in FIG. **20**.

(174) In addition, the n-type semiconductor region **101h** is formed at a central portion thereof with a p-type semiconductor region **201** for connection with a layer **202** formed at a silicon surface. The p-type semiconductor region **201** and the layer **202** are set at the same potential, which is different from the potential of the anode **105**.

(175) For example, the layer **202** may be set to the same potential as a ground (GND) or the cathode. In addition, by being configured to be at the same potential as the ground (GND) or the cathode, the layer **202** functions also as a hole accumulation region.

(176) In this way, the hole accumulation region (layer **202**) is formed in a region other than the cathode region at the surface of the silicon substrate (other than the projected portion **101h** of the n-type semiconductor region **101h**), and the cathode constituting a high-electric-field region is embedded in the inside of silicon.

(177) For example, like in the APD **21f** depicted in FIG. **15** in the sixth embodiment, of the region constituting the high-electric-field region, other portions than the portion connected with the contact of the cathode are embedded in the silicon substrate. In addition, the layer **202** is formed at the surface of the silicon substrate, in a substantially central area of the n-type semiconductor region **101h**, in a region (a region surrounded by the projected portion **101h**) between the portions (projected portion **101h**) connected with the wiring for the cathode. Further, a voltage (the same potential as the GND or the cathode) is impressed on the layer **202**, whereby the layer **202** is caused to function as a hole accumulation region.

(178) By such a configuration, the amount of damage due to inflow of a dark current generated at

the silicon surface and formation of a high-electric-field region can be reduced, and DCR can be suppressed. Besides, in the case where the layer **202** is set to the same potential as the cathode, circuit configuration can be simplified without increasing voltage systems.

(179) In the APD **21h** in the eighth embodiment, also, like in the first to seventh embodiments, there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(180) In addition, while an example in which the eighth embodiment is applied to the first embodiment (FIG. 3) has been depicted here, the eighth embodiment, specifically, the configuration in which the n-type semiconductor region **101** having the projected portion is formed and the region surrounded by the projected portion is covered with the layer **202** of the opposite conductivity type, may be applied to the second to seventh embodiments.

Configuration of APD in Ninth Embodiment

(181) FIGS. **21A** and **21B** depict sectional configurations in a ninth embodiment of the APD **21**. Of an APD **21i** illustrated in FIGS. **21A** and **21B**, the parts similar to those of the APD **21a** depicted in FIG. 3 are denoted by the same reference symbols as used above, and descriptions of them are omitted.

(182) The APD **21i** illustrated in FIG. **21A** is the same in structure as the APD **21h** depicted in FIG. **19**. The configuration in a section of the APD **21i** in the ninth embodiment is the same as that of the APD **21h** depicted in FIG. **19**, but the configuration in another section is different from that of the APD **21h** depicted in FIG. **19**, as illustrated in FIG. **21B**.

(183) While the APD **21h** in the eighth embodiment above has been an example in which the projected portion **101h** of the n-type semiconductor region **101h** is formed continuously (formed in a predetermined shape, for example, a tetragonal shape), the APD **21i** in the ninth embodiment has a configuration in which a projected portion **101i** of an n-type semiconductor region **101i** is formed discontinuously.

(184) FIG. **22** is a plan view of the APD **21i**. Referring to the APD **21i-1**, for example, depicted in FIG. **22**, the projected portion **101i** of the n-type semiconductor region **101i** is formed only at part of the inside of the anode **105**. In the example depicted in FIG. **22**, the projected portions **101i** are formed at four parts. The projected portions **101i** at the four parts are each formed at a substantially central portion of an edge (a side) of the n-type semiconductor region **101i**.

(185) Note that a case in which the projected portions **101i** of the n-type semiconductor region **101i** are each formed at a substantially central portion of an edge (a side) of the n-type semiconductor region **101i** has been taken as an example here, the projected portions **101i** may each be formed at part of the n-type semiconductor region **101i**, specifically, in a region other than the central portion of an edge (a side), for example, in a corner portion of the n-type semiconductor region **101i**.

(186) Of the APD **21i** having the configuration depicted in FIG. **22**, a sectional view taken along line A-A' is as depicted in FIG. **21A**, and a sectional view taken along line B-B' is as depicted in FIG. **21B**.

(187) The APD **21i** in the ninth embodiment, like the APD **21h** in the eighth embodiment, has a configuration in which the hole accumulation region (layer **202**) is formed in a region other than the cathode region (in this case, other than the projected portion **101i** of the n-type semiconductor region **101i**) of a surface of a silicon substrate, and the cathode constituting a high-electric-field region is embedded in the inside of silicon.

(188) By such a configuration, the amount of damage due to inflow of a dark current generated at a silicon surface and formation of a high-electric-field region can be reduced, and DCR can be suppressed. Besides, in the case where the layer **202** is set to the same potential as that of the cathode, circuit configuration can be simplified without increasing voltage systems.

(189) In addition, in the APD **21i** in the ninth embodiment, the part of the projected portion **101i** is smaller than the part of the projected portion **101h** of the APD **21h** in the eighth embodiment, so

that damage defects can be further reduced, more relaxation of electric field can be realized, and DCR can be further suppressed.

(190) In the APD **21i** in the ninth embodiment, also, like in the first to eighth embodiments, there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

(191) Besides, while an example in which the ninth embodiment is applied to the first embodiment (FIG. **3**) has been depicted here, the ninth embodiment, specifically, the configuration in which the n-type semiconductor region **101** having the projected portion is formed, the region surrounded by the projected portion is covered with the layer **202** of the opposite conductivity type, or the potential of the layer **202** is maintained at a predetermined potential, may be applied to the second to seventh embodiments.

Configuration of APD in Tenth Embodiment

(192) FIG. **23** depicts a sectional configuration in a tenth embodiment of the APD **21**. An APD **21j** illustrated in FIG. **23** is different in polarity from, but is the same in configuration as, the APDs **21** in the first to ninth embodiments.

(193) The APD **21j** depicted in FIG. **23** has the same configuration as, but is configured to be different in polarity from, the APD **21a** in the first embodiment depicted in FIG. **3**.

(194) The APD **21j** depicted in FIG. **23** includes a p-type semiconductor region **101j** the conductivity of which is p type, and an n-type semiconductor region **102j** which is formed under the p-type semiconductor region **101j** and the conductivity of which is n type. The p-type semiconductor region **101j** and the n-type semiconductor region **102j** are formed in a well layer **103j**.

(195) The well layer **103j** may be a semiconductor region of n-type conductivity, or may be a semiconductor region of p-type conductivity.

(196) The p-type semiconductor region **101j** functions as an anode, and is connected to a circuit through the contact **104**. A cathode **105j** opposed to the anode is formed between the p-type semiconductor region **101j** and the isolation region **108**, in the same layer as the p-type semiconductor region **101j**. The cathode **105j** is connected to a circuit through the contact **106**.

(197) Between the isolation region **108** and the well layer **103j** is formed an electron accumulation region **107j**, which may be formed as an n-type semiconductor region. The electron accumulation region **107j** is formed on the lower side of the cathode **105j**, and is formed in the state of being electrically connected with the cathode **105j**. In addition, the electron accumulation region **107j** is formed between the well layer **103j** and the isolation region **108**, and on the lower side (on the back surface side of the APD **21j**) of the well layer **103j**.

(198) In the case of the APD **21j** in which polarity is thus reversed, holes are trapped in the electron accumulation region **107j**. In the case where holes are trapped, also, DCR can be suppressed, like in the case where electrons are trapped.

(199) To the APD **21j** in which the polarity is thus reversed, also, the first to ninth embodiments can be applied. The descriptions of the cases where the first to ninth embodiments are applied and the polarity is reversed are similar to those of the above-described cases, and, accordingly, detailed description thereof is omitted here.

(200) The APD **21j** illustrated in FIG. **23** is an example in which the polarity of the APD **21a** in the first embodiment depicted in FIG. **3** is reversed. The p-type semiconductor region **101j** of the APD **21j** may be formed in a tetragonal shape in plan-view layout, like the n-type semiconductor region **101** of FIG. **4**, or may be formed in a circular shape in plan-view layout, like the n-type semiconductor region **101** of FIG. **5**.

(201) In addition, the electron accumulation region **107j** of the APD **21j** depicted in FIG. **23** may not be formed ranging to a depth portion of the silicon substrate but may be formed only at part of the inside of the isolation region **108** on the p-type semiconductor region **101j** side, like the n-type

semiconductor region **101** in the second embodiment (FIG. 6).

(202) Besides, like the APD **21c** depicted in the third embodiment (FIGS. 7 and 8), the APD **21j** may be formed with a fixed charge film **151j**, as depicted in FIG. 24. In the APD **21j** with the reversed polarity, the fixed charge film **151j** is a positive fixed charge film.

(203) In addition, like the APD **21d** depicted in the fourth embodiment (FIG. 9), the APD **21j** may be formed with a metallic film **171** in the isolation region **108**, though not illustrated. Besides, an electron accumulation layer may also be formed, by forming the metallic film **171** and impressing a voltage on the metallic film **171**.

(204) Besides, like the APD **21e** depicted in the fifth embodiment (FIG. 12), the APD **21j** may have a configuration in which the well layer **103j** is deep, and the electron accumulation region **107j** and the fixed charge film **151j** are combined with each other, though not illustrated. Further, the metallic film **171** may be combined with this. In addition, the electron accumulation region **107j** and the metallic film **171** may be combined with each other.

(205) In addition, like the APD **21f** depicted in the sixth embodiment (FIG. 15), the APD **21j** may have a configuration in which the p-type semiconductor region **101j** is shaped to have a projected portion in a substantially central area, and only a region connected to the contact **104** of the p-type semiconductor region **101j** is exposed to the surface of the silicon substrate, though not illustrated.

(206) Besides, like the APD **21g** depicted in the seventh embodiment (FIG. 17), the APD **21j** may have a configuration in which the region where the cathode **105j** is formed is reduced (a configuration in which the cathode **105j** is formed only at part of the inside of the isolation region **108**), though not illustrated.

(207) In addition, like the APD **21h** depicted in the eighth embodiment (FIG. 19), the APD **21j** may have a configuration in which the p-type semiconductor region **101j** is shaped to have a projected portion near an end, only the region connected to the contact **104** of the p-type semiconductor region **101j** is exposed to the surface of the silicon substrate, and a layer **202j** is formed in a central area, though not illustrated. In this case, the layer **202j** is set to the same potential as that of the ground (GND) or the anode.

(208) Besides, like the APD **21i** depicted in the ninth embodiment (FIGS. 21A and 21B), the APD **21j** may have a configuration in which the p-type semiconductor region **101j** is shaped to have a projected portion, only the region connected to the contact **104** of the p-type semiconductor region **101j** is exposed to the surface of the silicon substrate, the projected portion is formed discontinuously, and a layer **202j** is formed in a central area, though not illustrated. In this case, the layer **202j** is set to the same potential as that of the ground (GND) or the anode.

(209) In this way, the first to ninth embodiments can also be applied to the APD **21** reversed in polarity. In this case, like in the first to ninth embodiments, there can be obtained such effects as a reduction of DCR, a reduction of crosstalk, an effect by which carriers can be easily collected into high-electric-field regions due to formation of lateral electric fields, and an effect by which PDE can be enhanced.

Configuration of APD in Eleventh Embodiment

(210) FIG. 25 depicts a sectional configuration in an eleventh embodiment of the APD **21**. An APD **21k** illustrated in FIG. 25 is the same in basic configuration as the APDs **21** in the first to tenth embodiments, but differs from those APDs **21** in that edge breakdown can be prevented (or alternatively, mitigated) without generating a lowering in PDE or an increase of dark signals.

(211) While description will be made here by taking as an example a case where a configuration capable of preventing (or alternatively, mitigating) edge breakdown without generating a lowering in PDE or an increase of dark signals is applied to the APD **21a** in the first embodiment depicted in FIG. 3, the configuration is also applicable to the second to tenth embodiments.

(212) In the configuration of the APD **21k** illustrated in FIG. 25, a block layer **301** is formed on the well layer **103** side of a side wall of a hole accumulation region **107k**. The block layer **301** is provided for blocking electrons that might reach an n-type semiconductor region **101** without

passing through the multiplication region. Here, the electrons which reach the n-type semiconductor region **101** without passing through the multiplication region will now be described. (213) Description will be made referring again to the APD **21a** depicted in FIG. 3, for example. In the APD **21a** depicted in FIG. 3, the p-type semiconductor region **102** constituting the multiplication region is formed to be smaller than the n-type semiconductor region **101**, for preventing (or alternatively, mitigating) edge breakdown. With such a configuration, edge breakdown can be prevented (or alternatively, mitigated), but PDE may be worsened.

(214) FIG. 26 depicts, in an enlarged form, an edge portion of the multiplication region of the APD **21a** depicted in FIG. 3. The p-type semiconductor region **102** is formed to be smaller than the n-type semiconductor region **101**, whereby generation of edge breakdown can be restrained. When the p-type semiconductor region **102** is formed to be smaller than the n-type semiconductor region **101**, an electric field is formed in the direction indicated by the arrow in the figure, in the vicinity of an edge of the multiplication region. Since a conduction electron **331** moves along the direction of the electric field, the conduction electron **331** does not move toward the multiplication region but passes by the edge of the multiplication region (passes on the outside of the p-type semiconductor region **102**), to reach the cathode (n-type semiconductor region **101**).

(215) The conduction electrons **331** reaching the n-type semiconductor region **101** in this way are not multiplied because they do not pass the multiplication region, so that a signal is not detected at the cathode. If there are such conduction electrons **331** that are not detected, PDE would be lowered.

(216) In view of this, as illustrated in FIG. 25, the block layer **301** is provided for reducing the number of the conduction electrons **331** which would reach the n-type semiconductor region **101** without passing through the multiplication region. Where the block layer **301** is thus provided, the conduction electrons **331** are caused to pass through the multiplication region. This will be described referring to FIG. 27.

(217) FIG. 27 depicts, in an enlarged form, an edge portion of the multiplication region of the APD **21k** depicted in FIG. 25. In the vicinity of the edge of the multiplication region, an electric field is formed in the direction indicated by the arrow in the figure, namely, in the direction toward the center of the APD **21k**.

(218) Since the conduction electron **331** moves along the direction of the electric field, the conduction electron **331** goes toward the center of the APD **21k**, and, as a result, moves toward the p-type semiconductor region **102** and, hence, toward the multiplication region. Therefore, the conduction electrons **331** pass through the multiplication region, to reach the cathode (n-type semiconductor region **101**).

(219) Thus, with the block layer **301** provided, a potential barrier is formed, whereby even those conduction electrons **331** which would not reach the multiplication region if the block layer **301** is absent can be made to reach the multiplication region, so that PDE can be prevented (or alternatively, mitigated) from being lowered. In addition, where the p-type semiconductor region **102** constituting the multiplication region is formed to be smaller than the n-type semiconductor region **101**, it is also possible to prevent (or alternatively, mitigate) edge breakdown.

(220) In the APD **21k** illustrated in FIG. 25, the block layer **301** is formed in a projecting shape at side walls on the well layer **103** side of the hole accumulation region **107k**. The block layer **301** can be formed as a p-type semiconductor region, like the hole accumulation region **107k**.

(221) The block layer **301** may be formed in substantially the same layer as the p-type semiconductor region **102**, as illustrated in FIG. 28A. In addition, as depicted in FIG. 28B, the block layer **301** may be formed at a position on the lower side in the figure than the p-type semiconductor region **102**, in other words, at a position nearer to the incidence surface side than the p-type semiconductor region **102** when viewed from the incidence surface (the lower surface in the figure) side. In FIG. 28B, an upper edge of the block layer **301** is spaced by a distance B from a lower edge of the p-type semiconductor region **102**.

(222) Thus, when the p-type semiconductor region **102** is taken as a reference, the block layer **301** is provided in the range from the same position as the p-type semiconductor region **102** to the position slightly spaced (the position spaced by the distance B) therefrom.

(223) Note that when the distance B is enlarged, namely, when the block layer **301** is largely spaced from the p-type semiconductor region **102**, the conduction electrons **331** may bypass the block layer **301** to reach the n-type semiconductor region **101**, which may lessen the effect of the block layer **301** as a potential barrier. Therefore, the distance B is set within such a range as to avoid such an unfavorable situation. Accordingly, as depicted in FIG. 27, for example, the distance B may be 0, and the block layer **301** may be formed in such a position that the lower edge of the p-type semiconductor region **102** and the upper edge of the block layer **301** are in the same position.

(224) As depicted in FIGS. 28A and 28B, the block layer **301** is formed at a deeper position (a deeper position in terms of a position on the opposite side from the cathode) than the depth of the multiplication region (the region including a boundary region between the n-type semiconductor region **101** and the p-type semiconductor region **102**).

(225) In addition, the block layer **301** may be equivalent to or different from the hole accumulation region **107k** in concentration. For example, the concentration of the block layer **301** may be approximately 1×10^{16} to $1 \times 10^{20} / \text{cm}^3$. Note that the optimum concentration for the block layer **301** depends on the size and structure of the APD **21k**, the impurity concentration of the wafer, etc.; therefore, the optimum concentration for the block layer **301** is set taking such factors into consideration.

(226) The block layer **301** may be formed such that an end portion of the p-type semiconductor region **102** and an end portion of the block layer **301** are substantially on the same line, when viewed vertically, as illustrated in FIG. 28C. For example, when the position of the end portion of the p-type semiconductor region **102** is assumed to be a position P0, then the block layer **301** may be formed such that the end portion of the block layer **301** is located around the position P0 (in the left-right direction in the figure).

(227) It is considered that when the block layer **301** is formed such that the end portion of the block layer **301** is located at the position P0, the block layer **301** can be formed such as to serve effectively as a potential barrier and to prevent (or alternatively, mitigate) the multiplication region from being narrowed.

(228) On the other hand, when the block layer **301** is so formed that the end portion of the block layer **301** is spaced from the position P0, the function of the block layer **301** as a potential barrier may be lowered. Therefore, the block layer **301** is so formed that its end portion is located as close as possible to the position P0, whereby a configuration can be realized in which the conduction electrons **331** can be blocked efficiently.

(229) Thus, the block layer **301** is formed in the vicinity of the end portion of the p-type semiconductor region **102**. Besides, with the block layer **301** formed in the vicinity of the end portion of the p-type semiconductor region **102**, more conduction electrons **331** can be guided into the multiplication region, and PDE can be enhanced thereby.

(230) A plan view of the APD **21k**, when the APD **21k** is viewed from the upper surface side (the upward side in FIG. 25, or the n-type semiconductor region **101** side), is as depicted in FIG. 4. Since the description referring to FIG. 4 has been made above, an overlapping description is omitted.

(231) The APD **21a** depicted in FIG. 4 is read (differently) as APD **21k**. Each of the APDs **21k** is isolated by the isolation region **108** formed in a grid pattern. The anode **105** is formed inside the isolation region **108**. The well layer **103** is formed between the anode **105** and the n-type semiconductor region **101**. The n-type semiconductor region **101** is formed in a central area of the APD **21k**.

(232) The hole accumulation region **107k** is formed inside the isolation region **108**, though not seen when viewed from the upper surface side. In other words, the hole accumulation region **107k** is

formed in substantially the same region as that of the anode **105**. In addition, although the block layer **301** is not seen when viewed from the upper surface side, the block layer **301** is formed such as to overlap with that portion of the well layer **103** which is located between the anode **105** and the n-type semiconductor region **101** and with part of the n-type semiconductor region **101**.

(233) While the n-type semiconductor region **101** depicted in FIG. **4** represents an example of being formed in a tetragonal shape, the n-type semiconductor region **101** may be formed in a circular shape as depicted in FIG. **5**.

(234) In the case where the n-type semiconductor region **101** is formed in a tetragonal shape as illustrated in FIG. **4**, the area of the multiplication region (n-type semiconductor region **101**) can be secured to be large, so that detection efficiency called PDE can be enhanced.

(235) In the case where the n-type semiconductor region **101** is formed in a circular shape as depicted in FIG. **5**, electric field concentration at edge portions can be restrained, so that unintentional edge breakdown can be suppressed.

(236) Meanwhile, the configuration in which the block layer **301** is formed thereby to prevent (or alternatively, mitigate) edge breakdown and to ensure that conduction electrons **331** tending to reach the n-type semiconductor region **101** without passing through the multiplication region are guided to pass through the multiplication region may be a configuration in which the hole accumulation region **107k** and the isolation region **108** are eliminated. For example, an APD **21k'** having a configuration as illustrated in FIG. **29** may be adopted.

(237) The APD **21k'** depicted in FIG. **29** has a configuration obtained by eliminating the hole accumulation region **107k** and the isolation region **108** from the APD **21k** depicted in FIG. **25**. The block layer **301** is formed at a deeper position than the multiplication region, in such a size as to cover at least a region corresponding to the difference between the size of the n-type semiconductor region **101** and the size of the p-type semiconductor region **102**.

(238) In the case where the APD **21k'** is configured in this way, a configuration in which adjacent APDs **21k'** share the block layer **301**, as illustrated in FIG. **30**, for example, may also be adopted. An APD **21k'-1** and an APD **21k'-2** depicted in FIG. **30** are adjacent APDs **21k'**, and the block layer **30** is formed in such a size as to extend over both the APD **21k'-1** and the APD **21k'-2**, and functions both as a potential barrier in the APD **21k'-1** and as a potential barrier in the APD **21k'-2**.

(239) In this way, the block layer **301** may be formed in such size and shape as to be shared by a plurality of APDs **21k'**.

(240) Note that, while a configuration of the APD **21k'** in which the hole accumulation region **107k** and the isolation region **108** are eliminated has been taken as an example in the above description, the present technology is applicable to other configurations than the configuration in which the hole accumulation region **107k** and the isolation region **108** are omitted. Thus, a configuration in which only either one of the hole accumulation region **107k** and the isolation region **108** is eliminated may be adopted.

(241) Thus, the APD **21k** in the eleventh embodiment is provided with the high-electric-field region (multiplication region) and with the block layer for blocking electrons from moving to the side of an end portion of the high-electric-field region, the block layer being provided at a position deeper than the high-electric-field region and on the side of the end portion of the high-electric-field region.

(242) The block layer is formed at a deeper position than the high-electric-field region.

(243) The hole accumulation region for trapping electrons is further provided, and the block layer is formed at a side wall of the hole accumulation region.

(244) The isolation region for isolation from neighboring pixels is further provided.

(245) Thus, the p-type semiconductor region **102** is formed to be smaller than the n-type semiconductor region **101**, whereby generation of edge breakdown can be restrained. In addition, the formation of the block layer **301** enables a configuration in which more conduction electrons **331** can be made to reach the multiplication region, so that PDE can be enhanced.

(246) Further, with the hole accumulation region **107k** formed, PDE can be further enhanced. Further, the formation of the isolation region **108** makes it possible to reduce crosstalk. In addition, in these configurations, DCR is not worsened, and the above-mentioned effects can be obtained.

(247) Note that, while an example in which the eleventh embodiment is applied to the first embodiment (FIG. **3**) has been depicted here, the eleventh embodiment, specifically, the configuration in which the block layer **301** is provided, is also applicable to the second to tenth embodiments.

Configuration of APD in Twelfth Embodiment

(248) FIG. **31** depicts a sectional configuration in a twelfth embodiment of the APD **21**. An APD **21m** illustrated in FIG. **31** is basically the same in configuration as the APD **21** in the first to tenth embodiment, but differs from the APD **21** in that edge breakdown can be prevented (or alternatively, mitigated) without generating a lowering in PDE or an increase of dark signals.

(249) In addition, the APD **21m** in the twelfth embodiment is similar to that in the eleventh embodiment in that edge breakdown can be prevented (or alternatively, mitigated) without generating a lowering in PDE or an increase of dark signals.

(250) The configuration of the APD **21m** depicted in FIG. **31** is similar to the configuration of the APD **21a** depicted in FIG. **3**, but is different in that the n-type semiconductor region **101** is formed in a region of a different impurity concentration. Descriptions of the similar points of the configuration of the APD **21m** depicted in FIG. **31** to those of the configuration of the APD **21a** depicted in FIG. **3** are omitted.

(251) The n-type semiconductor region **101** includes an n-type semiconductor region **101-1** having a concentration (referred to as concentration n_1) comparable to that in the above-mentioned embodiments and an n-type semiconductor region **101-2** having a concentration (referred to as concentration n_2) lower than the concentration n_1 .

(252) For example, the concentration n_2 of the n-type semiconductor region **101-2** may be set a concentration of approximately 0.1 to 0.7 times the concentration n_1 of the n-type semiconductor region **101-1**. Note that optimum values for the concentration n_1 and the concentration n_2 depend on the size and structure of the APD **21m**, impurity concentration of the wafer, etc., and, therefore, the optimum values for the concentration n_1 and the concentration n_2 are individually set taking such factors into consideration.

(253) The n-type semiconductor region **101-2** is formed at both ends of the n-type semiconductor region **101**. Specifically, referring to FIG. **32**, when the APD **21m** is viewed from the upper surface side, the n-type semiconductor region **101-2** is formed such as to surround the periphery of the n-type semiconductor region **101-1**. In other words, the n-type semiconductor region **101-1** of the concentration n_1 is formed in a central area of the APD **21m**, and the n-type semiconductor region **101-2** of the concentration n_2 is formed in the periphery (outer peripheral area) thereof.

(254) Besides, though not seen when the APD **21m** is viewed from the n-type semiconductor region **101** side, the p-type semiconductor region **102** is formed on the lower side of the n-type semiconductor region **101**. The n-type semiconductor region **101-1** and the p-type semiconductor region **102** may be substantially the same in size. Alternatively, a configuration may be adopted in which the n-type semiconductor region **101-1** is formed as a region smaller than the p-type semiconductor region **102**, such that the n-type semiconductor region **101-2** overlaps with part (end portions) of the p-type semiconductor region **102**.

(255) In addition, as depicted in FIG. **33**, the n-type semiconductor region **101-1** and the n-type semiconductor region **101-2** may not be the same in thickness. Specifically, as illustrated in FIG. **33**, the thickness of the n-type semiconductor region **101-2** may be smaller than the thickness of the n-type semiconductor region **101-1**.

(256) Thus, the n-type semiconductor region **101** constituting the multiplication region includes two different patterns, namely, a pattern covering the multiplication region (n-type semiconductor region **101-1**) and a comparatively smaller pattern (n-type semiconductor region **101-2**), the two

different patterns being different in the concentration of an n-type impurity used for doping, whereby a configuration can be realized in which edge breakdown can be prevented (or alternatively, mitigated) without generating a lowering in PDE or an increase of dark signals.

(257) FIG. 34 depicts, in an enlarged form, an edge portion of the multiplication region in the APD 21m depicted in FIG. 31. When the n-type semiconductor region 101 includes the two regions that differ in concentration, an electric field is weaker in the vicinity of the n-type semiconductor region 101-2 than in the n-type semiconductor region 101-1. In other words, the electric field is relatively stronger in a central area (n-type semiconductor region 101-1) of the n-type semiconductor region 101 than in an outer peripheral area (n-type semiconductor region 101-2) of the n-type semiconductor region 101.

(258) Since a relative difference in electric field can be generated, conduction electrons 331 present in the vicinity of the n-type semiconductor region 101-2, or in the vicinity of an edge of the multiplication region, are drawn toward a relatively stronger electric field. In other words, the conduction electrons 331 present in the vicinity of the edge of the multiplication region can be guided to the multiplication region, so that more conduction electrons 331 can be made to reach the multiplication region.

(259) Thus, the APD 21m in the twelfth embodiment has a high-electric-field region (multiplication region) including a first layer and a second layer, and the first layer includes a first region having a first concentration and a second region having a second concentration.

(260) In addition, the second region is formed in an outer peripheral area of the high-electric-field region.

(261) Besides, the first concentration is set higher than the second concentration.

(262) In the APD 21m having such a configuration, an n-type semiconductor region of the multiplication region can be formed by use of an n-type impurity doping mask for forming the multiplication region and a p-type impurity doping mask for forming the multiplication region.

(263) Specifically, for example, using the n-type impurity doping mask for forming the multiplication region, the n-type semiconductor region 101 of the concentration n2 is formed, whereby a region corresponding to the n-type semiconductor region 101-2 can be formed.

Thereafter, using the p-type impurity doping mask for forming the multiplication region, doping with an n-type impurity is further applied to a region corresponding to the n-type semiconductor region 101-1, whereby the concentration of the n-type impurity is enhanced, and the n-type semiconductor region 101-1 of the concentration n1 can be formed.

(264) Therefore, even in the case of forming the n-type semiconductor region 101 including regions of different concentrations, the formation can be carried out using a comparable number of masks to the number of masks used in forming a conventional APD 21.

(265) Thus, with the p-type semiconductor region 102 formed to be smaller than the n-type semiconductor region 101, generation of edge breakdown can be restrained. In addition, with the n-type semiconductor region 101 formed by forming two regions of different concentrations, more conduction electrons 331 can be made to reach the multiplication region, so that PDE can be enhanced.

(266) Further, with a hole accumulation region 107m formed, PDE can be further enhanced. Note that though not illustrated, the APD 21m in the twelfth embodiment may be provided with a configuration in which the hole accumulation region 107m is eliminated, like the APD 21k in the eleventh embodiment.

(267) Furthermore, with the isolation region 108 formed, crosstalk can be reduced. Note that though not illustrated, the APD 21m in the twelfth embodiment may be provided with a configuration in which the isolation region 108 is eliminated, like the APD 21k in the eleventh embodiment. Note that though not illustrated, the APD 21m in the twelfth embodiment may be provided with a configuration in which the hole accumulation region 107m and the isolation region 108 are eliminated, like the APD 21k in the eleventh embodiment.

(268) In addition, in these configurations, the above-mentioned effects can be obtained, without worsening of DCR.

(269) Note that while an example in which the twelfth embodiment is applied to the first embodiment (FIG. 3) has been described here, the twelfth embodiment, specifically, the configuration in which the n-type semiconductor region **101** having the two regions of different concentrations is provided, may be applied to the second to tenth embodiments.

(270) Besides, the APD **21** can be produced as an APD having both the block layer **301** in the eleventh embodiment and the n-type semiconductor region **101** that has the two regions of different concentrations in the twelfth embodiment.

(271) The above-mentioned fixed charge film **151** (inclusive of **151e** and **151j**) includes, for example, a hafnium oxide (HfO₂) film, an aluminum oxide (Al₂O₃) film, a zirconium oxide (ZrO₂) film, a tantalum oxide (Ta₂O₅) film, or a titanium oxide (TiO₂) film. The just-mentioned kinds of films have actually been used as gate insulating films of insulated gate field effect transistors, etc., and film forming methods therefor have been established; accordingly, these films can be easily formed. Examples of the film forming method include a chemical vapor deposition method, a sputtering method, and an atomic layer deposition method. By use of the atomic layer deposition method, a SiO₂ layer for reducing interface level can be simultaneously formed in a thickness of approximately 1 nm during film formation, which is preferable.

(272) In addition, examples of other materials than the above-mentioned include lanthanum oxide (La₂O₃), praseodymium oxide (Pr₂O₃), cerium oxide (CeO₂), neodymium oxide (Nd₂O₃), promethium oxide (Pm₂O₃), samarium oxide (Sm₂O₃), europium oxide (Eu₂O₃), gadolinium oxide (Gd₂O₃), terbium oxide (Tb₂O₃), dysprosium oxide (Dy₂O₃), holmium oxide (Ho₂O₃), erbium oxide (Er₂O₃), thulium oxide (Tm₂O₃), ytterbium oxide (Yb₂O₃), lutetium oxide (Lu₂O₃), and yttrium oxide (Y₂O₃). Further, the above-mentioned fixed charge film **151** having a negative fixed charge may include a hafnium nitride film, an aluminum nitride film, a hafnium oxynitride film or an aluminum oxynitride film.

(273) Besides, the fixed charge film **151** may have silicon (Si) or nitrogen (N) added into the film in such a range as not to spoil insulating properties. The concentration of silicon (Si) or nitrogen (N) is appropriately determined in such a range as not to spoil the insulating properties of the film. With silicon (Si) or nitrogen (N) thus added, heat resistance of the film and ion injection inhibiting capability during the process can be enhanced.

(274) <In Regard of Configuration Inclusive of Peripheral Region>

(275) The APD **21** has been described in the above embodiments. As depicted in FIGS. 35 and 36, the APDs **21** are disposed in an array in a pixel region **A1** provided in a sensor chip **310**. In FIG. 36, an example is depicted in which an APD **21-1** and an APD **21-2** are disposed in an aligned state in the pixel region **A1**.

(276) A logic chip **610** is connected to a lower surface (a surface on the side opposite to a light incidence surface) of the sensor chip **310** on which the APDs **21** are disposed. The logic chip **610** is formed with a circuit that processes signals from the APDs **21** and supplies electric power to the APDs **21**.

(277) A peripheral region **A2** is disposed on the outside of the pixel region **A1**. Further on the outside of the peripheral region **A2**, a pad region **A3** is disposed.

(278) As depicted in FIG. 36, in the pad region **A3**, pad openings **313** which are vertical holes extending from an upper end of the sensor chip **310** to reach an inside of a wiring layer **311** and which are holes for wiring to an electrode pad **312** are formed to be aligned in a straight line.

(279) The electrode pad **312** for wiring is provided at a bottom of the pad opening **313**. The electrode pad **312** is used, for example, when being connected to a wiring in the wiring layer **311** or when being connected to other external device (chip or the like). In addition, a configuration may be adopted in which a wiring layer near an adhesion surface between the sensor chip **310** and the

logic chip **610** functions also as the electrode pad **312**.

(280) The wiring layer **311** formed in the sensor chip **310** and the wiring layer formed in the logic chip **610** each include an insulating film and a plurality of wirings. The plurality of wirings and the electrode pad **312** include a metal such as, for example, copper (Cu) or aluminum (Al). Wirings formed in the pixel region **A1** and the peripheral region **A2** also include a similar material.

(281) The peripheral region **A2** is provided between the pixel region **A1** and the pad region **A3**. While the configuration of the peripheral region **A2** will be described later, the peripheral region **A2** includes an n-type semiconductor region **321** and a p-type semiconductor region **322**. In addition, the p-type semiconductor region **322** is connected to a wiring **324** through a contact **325**, and the wiring **324** is connected to a ground (GND).

(282) In the example depicted in FIG. **36**, in the pixel region **A1**, the sensor chip **310** and the logic chip **610** are electrically connected in a form in which parts of those wiring layers formed on the side of the adhesion surface between the sensor chip **310** and the logic chip **610** which are located on the most adhesion surface side are directly bonded to each other.

(283) In the n-type semiconductor regions **321**, there are formed two trenches, namely, a trench **323-1** and a trench **323-2**. The trenches **323** are provided for securely isolating the pixel region **A1** and the peripheral region **A2**. FIG. **36** depicts a case in which two trenches **323** are formed. It is sufficient, however, that at least one trench **323** is formed, as will be described later referring FIG. **51** and subsequent figures.

(284) In the APD **21**, as has been described above, a high voltage is impressed between a cathode (contact **104**) and an anode **105**. Besides, the peripheral region **A2** is grounded. For this reason, in an isolation region provided between the pixel region **A1** and the peripheral region **A2**, a high-electric-field region may be generated due to the application of a high voltage to the anode **105**, and breakdown may be generated. Although it may be contemplated to broaden the isolation region provided between the pixel region **A1** and the peripheral region **A2** for the purpose of avoiding the breakdown, broadening the isolation region enlarges the size of the sensor chip **310**.

(285) For preventing (or alternatively, mitigating) such breakdown, the trenches **323** are formed. The trenches **323** makes it possible to prevent (or alternatively, mitigate) the breakdown, without broadening the isolation region. The trenches **323** will be described later referring to FIG. **51** and subsequent figures.

First Embodiment of Isolation Region

(286) The APDs **21** are formed with the isolation region for isolation between the APDs **21**. For instance, in the APD **21a** depicted in FIG. **3**, an isolation region **108** is formed.

(287) A detailed configuration of the isolation region **108** will be described below. The isolation region **108** described below will be described taking as an example an isolation region which is provided between the APD **21** disposed in an outermost peripheral area of the pixel region **A1** and the peripheral region **A2**.

(288) FIG. **37** is a figure depicting a configuration of an isolation region **108a** in the first embodiment. In the following description, the light incidence surface is illustrated on the upper side. Besides, in the following description, the APD **21** appearing in the description is one of the APDs **21a** to **21m** in the first to twelfth embodiments described above.

(289) On the light incidence surface side of the APD **21**, an on-chip lens **23** is formed. A planarizing film **401** is formed between the on-chip lens **23** and the APD **21**.

(290) Isolation regions **108** are formed at both ends of a well layer **103** of the APD **21**. Between the APDs **21** on the left side in the figure, the isolation region **108** includes a p-type semiconductor region **411**, a trench **412**, and a p-type semiconductor region **413**.

(291) In addition, between the APD **21** on the right side in the figure and the peripheral region **A2**, the isolation region **108** includes the p-type semiconductor region **411**, the trench **412**, the p-type semiconductor region **413**, an n-type semiconductor region **414**, and a p-type semiconductor region **415**. An n-type semiconductor region **416** is a part forming the peripheral region **A2**.

(292) Note that an anode **105** (not depicted in FIG. **40**) is formed between the p-type semiconductor region **411** and a contact **106**. In addition, while the p-type semiconductor region **411** is described to be included in the isolation region **108** for convenience of explanation, the p-type semiconductor region **411** can be a region corresponding, for example, to the above-mentioned hole accumulation region **107a** in FIG. **3**.

(293) As depicted in FIG. **37**, the isolation region **108a** is a region in which the trench **412** is formed between the p-type semiconductor region **411** and the p-type semiconductor region **413**.

(294) For instance, the p-type semiconductor region **411** and the p-type semiconductor region **413** are a single p-type semiconductor region, the p-type semiconductor region is bored so as to penetrate from the front surface to the back surface, and an insulating film such as an oxide film or a nitride film is formed in the bore (through-hole). For example, at the time of forming the planarizing film **401**, the material of the planarizing film **401** may be placed to fill the through-hole, whereby the trench **412** may be filled with a predetermined material. In this case, the planarizing film **401** and the trench **412** include the same material.

(295) A light-shielding film **402** is formed on an upper portion (light incidence surface side) of the trench **412**. The light-shielding film **402** includes a conductor such as a metal. The light-shielding film **402** is formed in an area ranging from an area on the p-type semiconductor region **415** to an area on the trench **412**. A contact **417** is formed on a lower surface (the side on which the light-shielding film **402** is not formed) of the p-type semiconductor region **415**, and is so configured that a predetermined voltage is impressed thereon.

(296) Here, the same voltage is impressed on the contact **106** connected to the anode **105** and on the contact **417**. The voltage applied to the contact **417** is applied also to the light-shielding film **402** through the p-type semiconductor region **415**. Since the light-shielding film **402** is formed also on the trench **412**, the same voltage as the voltage impressed on the contact **417** is impressed also on an upper portion of the trench **417**.

(297) With the light-shielding film **402** thus provided on the light incidence surface side and with the voltage thus impressed on the light-shielding film **402**, pinning in the vicinity of the light-shielding film **402** can be secured. In addition, in applying the voltage on the light-shielding film **402**, the voltage can be applied to the light-shielding film **402** through the p-type semiconductor region **415**, even in the case in which the contact **417** is provided on the surface (the surface opposite to the light incidence surface side) different from the side on which the light-shielding film **402** is formed.

(298) Therefore, the contacts (in FIG. **37**, the contact **104**, the contact **106**, and the contact **417**) can be formed on the same surface, so that connection thereof with wirings in the wiring layer **311** is facilitated.

(299) The trench **412** may be formed in part of the p-type semiconductor region, as depicted in FIG. **38**, without penetrating the p-type semiconductor region. In the sensor chip **310** illustrated in FIG. **38**, the trench **412'** is formed on the light incidence surface side in the p-type semiconductor region **411**, to extend to an intermediate portion of the p-type semiconductor region **411**.

(300) For instance, in a step prior to formation of the on-chip lens **23**, boring is conducted from the light incidence surface side of the APD **21**, to form the trench **412'**. In the case of the penetrating trench **412** depicted in FIG. **37**, deep boring is needed when the p-type semiconductor region **411** becomes deep; on the other hand, in the case of the trench **412'** depicted in FIG. **38**, deep boring into the p-type semiconductor region **411** is not needed, so that the trench **412'** can be formed more easily than the trench **412** depicted in FIG. **37**.

(301) In addition, even in the case of the trench **412'** as depicted in FIG. **38**, pinning in the vicinity of the light-shielding film **402** can be secured, since a voltage can be applied to the light-shielding film **402**.

(302) In this way, the depth of the trench **412** may be so set that the trench **412** penetrates the p-type semiconductor region **411**, or may be so set that the trench **412** is formed in part of the region

of the p-type semiconductor region **411**.

Second Embodiment of Isolation Region

(303) A second embodiment of the isolation region will be described referring to FIG. **39**. The isolation region **108a** in the first embodiment of the isolation region described above has been described taking as an example a case in which the light-shielding film **402** is used as a wiring for connecting the p-type semiconductor region **415** and the trench **412**. An isolation region **108b** in the second embodiment of the isolation region differs from the isolation region **108a** in the first embodiment of the isolation region, in that a wiring is formed to connect the p-type semiconductor region **415** and the trench **412**.

(304) Referring to FIG. **39**, a contact **451-1** is formed on the light incidence surface side of the p-type semiconductor region **415**, and is connected to a wiring **452**. In addition, a contact **451-2** is formed on a light-shielding film **402'** on the light incidence surface side of the trench **412**, and is connected to the wiring **452**. Therefore, the p-type semiconductor region **415** and the trench **412** are connected to each other through the contact **451-1**, the wiring **452**, and the contact **451-2**.

(305) Also in such a case, the voltage impressed on the contact **417** is impressed also on the p-type semiconductor region **415**, the contact **451-1**, the wiring **452**, and the contact **451-2**, and is impressed also on the trench **412**. Therefore, like in the first embodiment of the isolation region, pinning in the vicinity of the contact **451-2** (the light-shielding film **402'**) can be secured.

(306) The configuration of the sensor chip **310** depicted in FIG. **38** can be applied to the configuration of the sensor chip **310** depicted in FIG. **39**, to obtain a configuration in which the trench **412** is formed to extend to an intermediate portion of the p-type semiconductor region **411**.

Third Embodiment of Isolation Region

(307) Referring to FIG. **40**, a third embodiment of the isolation region will be described. In the first and second embodiments of the isolation region, the material placed to fill the trench **412** has been described to be, for example, the same material as that of the planarizing film **401** and to be, for example, an insulating material.

(308) The trench **412** may be filled with a conductive material. Hereinafter, the conductive material placed to fill the trench **412** will be referred to appropriately as light-shielding wall. The light-shielding wall **471** can include the same material as that of the light-shielding film **402**.

Alternatively, the light-shielding wall **471** may include a material different from that of the light-shielding film **402**.

(309) The light-shielding wall **471** includes, for example, a metallic material such as W (tungsten), Al (aluminum), Cu (copper), etc. or such a material as polysilicon.

(310) In addition, a contact portion between the light-shielding wall **471** and the p-type semiconductor region **411** (the p-type semiconductor region **413**) is insulated by SiO₂ (silicon dioxide) or SiN (silicon nitride). Specifically, the trench **412** has a configuration in which a central portion includes the light-shielding wall **471**, and an insulating film is formed around the light-shielding wall **471**.

(311) According to the configuration of the sensor chip **310** illustrated in FIG. **40**, a voltage impressed on the contact **417** is impressed also on the p-type semiconductor region **415**, and is impressed also on the light-shielding film **402** in contact with the p-type semiconductor region **415**. Further, since the light-shielding film **402** is in connection with the light-shielding wall **471** in the trench **412**, the voltage is impressed also on the light-shielding wall **471**.

(312) Here, the voltage value of the voltage impressed on the contact **417** is assumed to be a voltage value A. The voltage impressed on the anode **105** (the voltage impressed on the contact **106**) is also assumed to be a voltage value A. In other words, the voltage applied to the contact **106** and the voltage applied to the contact **417** are assumed to be the same voltage. With the voltages thus made to be the same voltage, no electric field stress is exerted between the p-type semiconductor region **411** adjacent to the light-shielding wall **471** and the trench **412**, and between the p-type semiconductor region **413** and the trench **412**, so that deterioration of these portions can

be prevented (or alternatively, mitigated).

(313) In addition, with the light-shielding wall **471** formed in the trench **412** and with the voltage impressed on the light-shielding wall **471**, pinning can be secured. This configuration is a configuration corresponding to the APD **21d'** depicted in FIG. **11**.

(314) Referring again to FIG. **11**, a metallic film **171'** is formed on one side surface of the isolation region **108**, and a voltage is impressed on the metallic film **171'**. With the voltage impressed on the metallic film **171'**, a hole accumulation region **107d'** (not depicted in FIG. **11**) is formed in the vicinity of the metallic film **171'**.

(315) The metallic film **171'** in the APD **21d'** depicted in FIG. **11** corresponds to the light-shielding wall **471** of the sensor chip **310** depicted in FIG. **40**. Therefore, with the voltage impressed on the light-shielding wall **471**, a hole accumulation region **107d'** (not depicted in FIG. **40**) is formed in the vicinity of the light-shielding wall **471**.

(316) Also in the case of the configuration in which the voltage is thus impressed on the light-shielding wall **471** to form the hole accumulation region **107d'** (not depicted) in the vicinity of the light-shielding wall **471**, electrons generated at an interface between the well layer **103** and an isolation region **108c** (the p-type semiconductor region **411**) can be trapped, so that a dark current generated at the interface can be reduced.

(317) The isolation region **108** is formed also between the APDs **21**. For example, as has been described referring to FIG. **4**, in the plan view as viewed from the light incidence surface side, the isolation region **108** is formed to surround the APD **21**. The isolation region **108c** depicted in FIG. **40** also is formed to surround the APD **21**, and, therefore, the trench **412** is also formed to surround the APD **21**.

(318) While the trenches **412** are illustrated as if they were formed at both ends of the APD **21** in FIG. **40**, the trenches **412** depicted at both ends are continuous with each other. Therefore, the light-shielding wall **471** formed in the trench **412** is also formed to surround the APD **21**. Besides, though not depicted, since the trench **412** is continuous also with the trenches **412** of the neighboring APDs **21**, the light-shielding walls **471** formed in the neighboring trenches **412** are also formed in a continuous state.

(319) In other words, the light-shielding wall **471** formed between the APDs **21** formed in the pixel region **A1** is formed in a wholly continuous state. Therefore, as depicted in FIG. **40**, when a voltage is impressed on the light-shielding wall **471** of the trench **412** in the isolation region **108c** disposed in the peripheral region **A2**, the same voltage can be impressed on the light-shielding wall **471** surrounding each APD **21** of all the APDs **21** formed in the pixel region **A1**.

(320) Accordingly, hole accumulation regions can be formed in all the APDs **21**, the electrons generated at the interface between the well layer **103** and the isolation region **108c** (the p-type semiconductor region **411**) can be trapped, and the dark current generated at the interface can be reduced.

(321) The trench **412** may be formed in part of the p-type semiconductor region, as depicted in FIG. **41**, without penetrating the p-type semiconductor region. In the isolation region **108c'** depicted in FIG. **41**, the trench **412'** is formed on the light incidence side in the p-type semiconductor region **411**, and is formed to extend to an intermediate portion of the p-type semiconductor region **411**. This point is similar to that in the isolation region **108a'** depicted in FIG. **38**. In the isolation region **108c'** depicted in FIG. **41**, a light-shielding wall **471'** is also formed, and the light-shielding wall **471'** is formed in the trench **412'** which is formed to extend to an intermediate portion of the p-type semiconductor region **411**.

(322) Even in the case of the trench **412'** and the light-shielding wall **471'** as depicted in FIG. **41**, a voltage can be impressed on the light-shielding wall **471'**, whereby pinning in the vicinity of the light-shielding wall **471'** can be secured.

(323) In this way, the depth of the trench **412** may be so set that the trench **412** penetrates the p-type semiconductor region **411**, or may be so set that the trench **412** is formed in part of the region

of the p-type semiconductor region **411**.

Fourth Embodiment of Isolation Region

(324) In the first to third embodiments of the isolation region, an example in which the trench **412** is formed has been described. With the trench **412** formed, isolation between pixels and between the pixel region **A1** and the peripheral region **A2** can be performed securely, and, by impressing a voltage on the light-shielding film **402** connected to the trench **412**, pinning can be secured.

(325) As a configuration in which pinning can be secured, there may also be applied a form in which no trench **412** is formed, as depicted in FIG. **42**. A sensor chip **310** depicted in FIG. **42** has a configuration in which the trench **412** is eliminated from the isolation region **108a** in the first embodiment of the isolation region depicted in FIG. **37**, for example.

(326) In the configuration of the isolation region **108d** depicted in FIG. **42**, the contact **417** is connected to the p-type semiconductor region **415**, the p-type semiconductor region **415** is connected to the light-shielding film **402**, and the light-shielding film **402** is connected to the p-type semiconductor region **411**. The p-type semiconductor region **411** is not provided with any trench **412**.

(327) Even in this case, when a predetermined voltage is impressed on the contact **417**, the voltage is impressed also on the light-shielding film **402**. Therefore, in the vicinity of that portion of the p-type semiconductor region **411** which is in contact with the light-shielding film **402**, pinning can be secured.

Fifth Embodiment of Isolation Region

(328) In the first to fourth embodiments of the isolation region, an example is described in which the p-type semiconductor region **415** and the trench **412** are connected through the light-shielding film **402** provided on the light incidence surface side, as seen, for example, when referring again to the isolation region **108a** depicted in FIG. **37**. A configuration in which the p-type semiconductor region **415** and the trench **412** are connected on the side of the wiring layer **311** will be described below, as a fifth embodiment of the isolation region.

(329) In an isolation region **108e** depicted in FIG. **43**, a light-shielding film **491** is formed on the light incidence surface side of the p-type semiconductor region **415**. Unlike the above-mentioned light-shielding film **402** (for example, FIG. **37**), the light-shielding film **491** is not formed to extend to the trench **412**. In other words, as depicted in FIG. **43**, the light-shielding film **491** is not connected to the trench **412**.

(330) In addition, the light-shielding film **491** is formed with a contact **492**, and a voltage is impressed on the contact **492**. The voltage impressed on the contact **492** is made to be the voltage impressed on the contact **417** (for example, FIG. **37**) in the above embodiments. In other words, in the fifth embodiment of the isolation region, the voltage is impressed from the light incidence surface side through the light-shielding film **491** and the contact **492** which are formed on the light incidence surface side.

(331) The voltage impressed on the light-shielding film **491** is impressed also on the p-type semiconductor region **415**, and is impressed also on the light-shielding wall **472** formed in the trench **412**, through contacts and a wiring which are formed in the wiring layer **311**.

(332) A contact **493** is formed on the wiring layer **311** side of the p-type semiconductor region **415**, and the contact **493** is connected to a wiring **494**. The contact **493** is connected to one end side of the wiring **494**, and a contact **495** is connected to the other end side. The contact **495** is connected to the light-shielding wall **472**.

(333) Since the light-shielding film **491**, the p-type semiconductor region **415**, the contact **493**, the wiring **494**, the contact **495**, and the light-shielding wall **472** are thus connected, a configuration can be realized in which the voltage impressed on the light-shielding film **491** is impressed also on the p-type semiconductor region **415**, the contact **493**, the wiring **494**, the contact **495**, and the light-shielding wall **472**.

(334) In this case, also, with the voltage impressed on the light-shielding wall **472**, a hole

accumulation region can be formed in the vicinity of the light-shielding wall **472**. With the hole accumulation region formed, electrons generated at the interface between the well layer **103** and the isolation region **108e** (the p-type semiconductor region **411**) can be trapped, so that a dark current generated at the interface can be reduced.

Sixth Embodiment of Isolation Region

(335) In the first to fifth embodiments of the isolation region, an anti-flare film for preventing (or alternatively, mitigating) light reflection (flare) on the light-shielding film may be formed on the light-shielding film **402** (light-shielding film **491**). Here, a case of forming an anti-flare film on the light-shielding film **402** in the third embodiment of the isolation region depicted in FIG. **40** is depicted in FIG. **44**, as a sixth embodiment of the isolation region, and description will be continued.

(336) In a sensor chip **310** depicted in FIG. **44**, an anti-flare film **501** is formed on the light-shielding film **402**. With the anti-flare film **501** formed, reflection from the light-shielding film **402** can be reduced.

(337) Further, as depicted in FIG. **45**, an anti-reflection film can be formed. Referring to FIG. **45**, an anti-reflection film **502** is formed on the light-shielding film **402** (the anti-flare film **501** formed on the light-shielding film **402**) and an on-chip lens **23**. With the anti-reflection film **502** thus formed, influences of reflection of light from the light-shielding film **402**, the light-shielding wall **471** and the like can be reduced.

(338) Further, as depicted in FIG. **46**, a color filter **511** can be formed. While an example in which the APD **21** is formed with a color filter **511** has not been described in any of the above embodiments, the APD **21** can be formed with a color filter **511** in any of the embodiments.

(339) As depicted in FIG. **46**, the color filter **511** is formed between the on-chip lens **23** and the planarizing film **401**. The color filter **511** may be, for example, a filter transmissive to a specific color such as red (R), green (G), blue (B), white (W) or the like, or may be a filter transmissive to infrared light or ultraviolet light or the like.

Seventh Embodiment of Isolation Region

(340) In the first to sixth embodiments of the isolation region, the p-type semiconductor region **411**, the trench **412**, the p-type semiconductor region **413**, the n-type semiconductor region **414**, and the p-type semiconductor region **415** are aligned, as seen, for example, when referring to the isolation region **108c** depicted in FIG. **40**. All these regions can be formed as p-type semiconductor regions.

(341) FIGS. **47** and **48** are figures depicting a configuration of an isolation region **108g** in a seventh embodiment of the isolation region. FIG. **47** illustrates a case in which the isolation region **108c** depicted in FIG. **40** includes only a p-type semiconductor region. FIG. **48** illustrates a case in which the isolation region **108f** depicted in FIG. **45** includes only a p-type semiconductor region.

(342) The isolation region **108g** depicted in FIGS. **47** and **48** includes a p-type semiconductor region **411**, a trench **412**, and a p-type semiconductor region **413'**. The p-type semiconductor region **413'** is formed in a region corresponding, for example, to the p-type semiconductor region **413**, the n-type semiconductor region **414**, and the p-type semiconductor region **415** in the isolation region **108c** depicted in FIG. **40**.

(343) A contact **417** is connected to the p-type semiconductor region **413'**. Therefore, like in the isolation region **108c** depicted in FIG. **40**, also in the isolation region **108g** depicted in FIG. **47**, when a voltage at a predetermined voltage value is impressed on the contact **417**, the voltage can be impressed also on the p-type semiconductor region **413'**, the light-shielding film **402**, and the light-shielding wall **471**.

(344) Also in this case, therefore, a configuration in which a voltage can be impressed on the light-shielding wall **471** can be obtained, and a hole accumulation region can be formed in the vicinity of the light-shielding wall **471**. With the hole accumulation region formed, electrons generated at the interface between the well layer **103** and the isolation region **108g** (the p-type semiconductor region **411**) can be trapped, so that a dark current generated at the interface can be reduced.

(345) Note that though not illustrated, the seventh embodiment concerning the configuration in which the above-mentioned regions are formed only of the p-type semiconductor region can be applied to isolation regions **108** other than the isolation region **108c** depicted in FIG. **40** and the isolation region **108f** depicted in FIG. **45**.

Eighth Embodiment of Isolation Region

(346) In the first to seventh embodiments of the isolation region, a configuration in which a voltage is impressed on the trench **412** through the p-type semiconductor region **413'** (**415**) has been taken as an example and described. A configuration can also be adopted in which a voltage is impressed directly on the light-shielding wall **471** formed in the trench **412**.

(347) FIG. **49** is a figure for explaining a configuration of an isolation region **108h** in an eighth embodiment of the isolation region. The trench **412** of the isolation region **108h** is formed with the light-shielding wall **471**. A wiring **531** is connected to the light-shielding wall **471**. The wiring **531** may be formed on the isolation region **108h** to function also as a light-shielding film, in the same manner as the light-shielding film **402** (for example, FIG. **48**).

(348) The wiring **531** is formed with a contact **532**. When a predetermined voltage is impressed on the contact **532**, the voltage is impressed also on the wiring **531** and the light-shielding wall **471**. Therefore, in this case, also, the voltage can be impressed on the light-shielding wall **471**, whereby a hole accumulation region is formed in the vicinity of the light-shielding wall **471**, and a dark current generated at the interface can be reduced.

(349) According to the configuration of the isolation region **108h** depicted in FIG. **49**, the voltage is not impressed on the light-shielding wall **471** through the p-type semiconductor region, and, therefore, the region formed under the wiring **531** is an n-type semiconductor region **414'**. In this case, the isolation region **108h** includes the p-type semiconductor region **411**, the trench **412**, the p-type semiconductor region **413**, and the n-type semiconductor region **414'**.

(350) A configuration can also be adopted in which the trench **412** is formed with the light-shielding wall **471**, and the voltage is not impressed on the light-shielding wall **471**. FIG. **50** is a figure illustrating a configuration of the isolation region **108h** in the case in which no voltage is impressed on the light-shielding wall **471**. An isolation region **108h'** depicted in FIG. **50** is similar in configuration to the isolation region **108h** depicted in FIG. **49**, except that the contact **532** is eliminated from the isolation region **108h** depicted in FIG. **49**.

(351) The light-shielding wall **471** in the isolation region **108h'** depicted in FIG. **50** functions as a light-shielding wall for preventing (or alternatively, mitigating) influences of stray light coming from the neighboring APDs **21**.

(352) In this way, a configuration in which a voltage is impressed on the light-shielding wall **471** is possible, and, also, a configuration in which no voltage is impressed on the light-shielding wall **471** is possible. In the case in which no voltage is impressed on the light-shielding wall **471**, an effect of shielding stray light coming from the neighboring APDs **21** can be obtained. In the case in which a voltage is impressed on the light-shielding wall **471**, an effect such that pinning can be secured can further be obtained.

(353) Thus, in the first to eighth embodiments of the isolation region obtained by application of the present technology, a voltage can be impressed on a desired part, for example, the trench **412** (FIG. **37**), even if a physical through electrode is not formed.

(354) For instance, referring again to FIG. **37**, at the time of impressing a voltage on the light-shielding film **402**, application of the voltage on the contact **417** results in that the voltage is impressed on the light-shielding film **402** through the p-type semiconductor region **415**. In this case, a configuration in which the voltage is impressed on the light-shielding film **402** can be realized by forming a through electrode in the p-type semiconductor region **415** and connecting the contact **417** and the light-shielding film **402** to each other. In the present embodiment, however, a configuration in which the voltage is impressed on the light-shielding film **402** can be realized even though a through electrode is not formed, as above-mentioned.

(355) In addition, in the case in which the light-shielding film **402** is formed and a voltage is impressed on the light-shielding film **402** so as thereby to secure pinning, it is unnecessary for a current to flow in the light-shielding film **402** or the trench **412**. In other words, a configuration in which pinning in the vicinity of the light-shielding film **402** can be secured even without a configuration in which a current flows from the light-shielding film **402** into the trench **412** through the planarizing film **401**.

(356) For this reason, it is unnecessary to secure ohmic contact at the connection portion between the light-shielding film **402** and the trench **412**, so that manufacturing steps can be simplified.

Configuration of Peripheral Region

(357) A configuration of the peripheral region **A2** formed between the pixel region **A1** (FIG. 35) and the pad region **A3** will be described below.

(358) A comparatively high voltage is constantly impressed on the APDs **21**. In other words, a comparatively high voltage is constantly impressed on the pixel region **A1**. On the other hand, in the peripheral region **A2** outside of the pixel region **A1**, it is necessary to suppress a voltage to GND (keep a reference voltage). For this reason, it is necessary to secure an isolating property between the pixels and the peripheral region and reliability thereof, against application of a high voltage.

(359) For instance, a high voltage impressed on the pixel region **A1** may generate a high-electric-field region in the isolation region, whereby breakdown may be generated. Therefore, a configuration in which such a breakdown would not be generated should be provided.

(360) In view of this, as will be described below, a trench is formed in the peripheral region **A2** to thereby secure an isolating property between pixels and the peripheral region and reliability thereof.

First Embodiment of Peripheral Region

(361) FIG. 51 is a figure illustrating a configuration of a peripheral region in a first embodiment of the peripheral region.

(362) FIG. 51 depicts an APD **21** and a peripheral region **701a**. As the APD **21**, one of the APDs **21a** to **21m** in the first to twelfth embodiments can be applied. In addition, as an isolation region **108** between the APD **21** and the peripheral region **701a**, one of the isolation regions **108a** to **108h** in the first to eighth embodiments of the isolation region can be applied.

(363) The peripheral region **701a** depicted in FIG. 51 includes an n-type semiconductor region **321** and a p-type semiconductor region **322**. The n-type semiconductor region **321** is formed therein with a trench **323**. A contact **325** is connected to the p-type semiconductor region **322**.

(364) A reference voltage is impressed on the contact **325** connected to the p-type semiconductor region **322** (the contact **325** is kept at GND), and the p-type semiconductor region **322** is also maintained at the reference voltage.

(365) The trench **323** formed in the n-type semiconductor region **321** is formed to penetrate from the light incidence surface side to the side of a wiring layer **311**. Inside the trench **323** is formed an insulating film such as an oxide film or a nitride film (the inside is filled with a material capable of forming such an insulating film). While an example in which the position of the trench **323** is substantially central to the n-type semiconductor region **321** is depicted in FIG. 51, the trench **323** may be formed in a position other than the central position.

(366) FIG. 52 is a plan view, as viewed from the light incidence surface side, of a sensor chip **310** in which one trench **323** is formed in the n-type semiconductor region **321** as depicted in FIG. 51. The pixel region **A1** is disposed in a central portion of the sensor chip **310**, and the peripheral region **A2** is formed around the pixel region **A1**. In this case, the peripheral region **A2** is the peripheral region **701a**.

(367) The trench **323** is formed in the peripheral region **701a**, and the trench **323** is also disposed continuously around the pixel region **A1**. In the example depicted in FIG. 52, a case in which the trench **323** is formed in a tetragonal shape is illustrated.

(368) The shape of the trench **323** is not limited to a tetragonal shape; as depicted in FIG. **53A**, the shape may be a shape as if obtained by cutting off angular portions (corner portions) of a tetragonal shape, with a circumference having a series of obtuse angles. Further, as depicted in FIG. **53B**, the shape may be a shape obtained by changing the angular portions of a tetragon into arcuate shapes so that the shape has no angular portion.

(369) With the trench **323** set in a shape free of acute angle portions, as depicted in FIGS. **53A** and **53B**, the trench **323** can be formed in a state in which its line width is maintained at a constant line width, whereby a filling-up property and an insulating property at corner portions can be enhanced.

(370) By thus forming the trench **323** in the peripheral region **701a**, in other words, by forming an insulating film in the peripheral region **701a** such as to surround the pixel region **A1**, an isolating property between the pixels and the peripheral region and reliability thereof can be secured.

(371) Further, by providing two trenches **323**, as depicted in FIG. **54**, the insulating property between the pixels and the peripheral region and reliability thereof can be secured more assuredly. Two trenches **323-1** and **323-2** are formed in the n-type semiconductor region **321** in a peripheral region **701a'** depicted in FIG. **54**.

(372) The number of the trenches **323** thus provided in the peripheral region **701** is not limited to two, and it is sufficient that the number is at least one. Further, though not depicted, three or more trenches **323** may be formed. With the number of the trenches **323** increased, the isolating property between the pixels and the peripheral region and reliability thereof can be secured more assuredly.

(373) FIG. **55** is a plan view, as viewed from the light incidence surface side, of a sensor chip **310** in which the two trenches **323-1** and **323-2** are formed in the n-type semiconductor region **321** as depicted in FIG. **54**. The peripheral region **A2** (peripheral region **701a'**) is formed around the pixel region **A1** disposed in a central portion of the sensor chip **310**.

(374) The trench **323-1** is formed in the peripheral region **701a'** on the side nearer to the pixel region **A1**, and the trench **323-2** is formed in the peripheral region **701a'** on the side farther from the pixel region **A1**. The trenches **323-1** and **323-2** are each formed continuously around the pixel region **A1**.

(375) While a case in which the trenches **323-1** and **323-2** are formed in a tetragonal shape is illustrated in the example depicted in FIG. **55**, the shape may be a shape in which corner portions are formed in an obtuse angle shape or an arcuate shape, like the trenches **323** depicted in FIG. **53A**.

(376) In the case in which a plurality of trenches **323** are formed in the n-type semiconductor region **321**, the materials placed to fill the plurality of trenches **323** (the materials of insulating films formed in the trenches **323**) may be the same material or may be different materials.

Second Embodiment of Peripheral Region

(377) A peripheral region **701** in a second embodiment of the peripheral region will be described referring to FIG. **56**. Note that in the following description, a case in which two trenches **323** are formed will be taken as an example.

(378) In a peripheral region **701b** illustrated in FIG. **56**, two trenches, namely, a trench **323-1** and a trench **323-2** are formed. In the trenches **323-1** and **323-2**, insulating films are formed, and light-shielding walls **751-1** and **751-2** including a metal or a conductive material are also formed.

(379) While the light-shielding walls **751** are provided for insulation between the pixel region **A1** and the peripheral region **A2** and are not walls provided for the purpose of shielding light, they can include a material similar to that of the above-mentioned light-shielding wall **472** (FIG. **43**) and can be produced by a similar production step; for this reason, they are referred to as light-shielding walls here. In addition, while the light-shielding walls **751** are provided for the purpose of insulation, as a result of the provision of the light-shielding walls **751**, influences of stray light coming from the peripheral region **A2** to the pixel region **A1** can be reduced, so that a light-shielding effect can also be obtained.

(380) The light-shielding walls **751** include, for example, a metallic material such as Ti (titanium),

TiN (titanium nitride), Ta (tantalum), TaN (tantalum nitride), W (tungsten), WN (tungsten nitride), Al (aluminum), Cu (copper), etc. or such a material as polysilicon.

(381) In addition, a contact part between the light-shielding wall **751** and the n-type semiconductor region **321** is insulated by SiO₂ (silicon dioxide) or SiN (silicon nitride). In other words, of the trench **323**, a central portion includes the light-shielding wall **751**, and the insulating film is formed therearound.

(382) With not only the insulating films but also the light-shielding walls **751** thus formed, the insulating property between the pixels and the peripheral region and reliability thereof can be secured more assuredly.

Third Embodiment of Peripheral Region

(383) A peripheral region **701** in a third embodiment of the peripheral region will be described referring to FIG. **57**.

(384) A peripheral region **701c** depicted in FIG. **57** differs from the peripheral region **701b** depicted in FIG. **56** in that STIs (Shallow Trench Isolations) **771** are added.

(385) An STI **771-1** is formed on the wiring layer **311** side of a trench **323-1** (light-shielding wall **751-1**), and an STI **771-2** is formed on the wiring layer **311** side of a trench **323-2** (light-shielding wall **751-2**). While the STI **771** is a technique at the time of forming an element isolation region, here it is used as a stopper at the time of forming the trench **323**.

(386) In the case of forming the trench **323** by boring from the light incidence surface side, the STI **771** is formed on the side which is on the wiring layer **311** side of the n-type semiconductor region **321**, after which boring is conducted using the thus formed STI **771** as a stopper at the time of boring. After the boring, formation of an insulating film and formation of the light-shielding wall **751** are performed.

(387) Thus, a configuration can be adopted in which the STIs **771** are formed in the peripheral region **701c**, and, by using the STIs **771** as a stopper at the time of forming the trenches **323**, the trenches **323** and the light-shielding walls **751** are formed.

(388) In the case of this configuration, also, the isolating property between pixels and the peripheral region and reliability thereof can be secured more assuredly, like in the above-mentioned embodiments.

Fourth Embodiment of Peripheral Region

(389) A peripheral region **701** in a fourth embodiment of the peripheral region will be described referring to FIG. **58**.

(390) A peripheral region **701d** depicted in FIG. **58** differs from the peripheral region **701a'** depicted in FIG. **54** in that a wiring **324'** obtained by extending the wiring **324** is provided, and a contact **791** is added.

(391) In the case in which a plurality of trenches **323** are formed, floating may occur between the trenches **323**. For preventing (or alternatively, mitigating) a floating state from occurring between the trenches **323**, such a configuration as the peripheral region **701d** depicted in FIG. **58** is provided.

(392) Referring to FIG. **58**, a contact **791** is formed on an n-type semiconductor region **321** located between a trench **323-1** and a trench **323-2**. The contact **791** is connected to a wiring **324'**. The wiring **324'** is connected also to a contact **325** which is connected to a p-type semiconductor region **322**.

(393) In such a configuration, the n-type semiconductor region **321** located between the trench **323-1** and the trench **323-2** and the p-type semiconductor region **322** are connected to each other through the wiring **324'**.

(394) Since a reference voltage is impressed on the wiring **324'**, the n-type semiconductor region **321** and the p-type semiconductor region **322** are at the same voltage, for example, 0 volt. The configuration in which the potential of the n-type semiconductor region **321** is fixed in this way makes it possible to prevent (or alternatively, mitigate) a floating state from occurring between the

trench **323-1** and the trench **323-2**.

Fifth Embodiment of Peripheral Region

(395) A peripheral region **701** in a fifth embodiment of the peripheral region will be described referring to FIG. **59**.

(396) A peripheral region **701e** depicted in FIG. **59** has a configuration in which light-shielding walls **751** are formed in trenches **323**, like the peripheral region **701b** depicted in FIG. **56**. The light-shielding walls **751** are formed using a conductive material, and a voltage is impressed thereon. A light-shielding wall **751-1** and a light-shielding wall **751-2** are connected to a wiring **811**. A contact **812** is connected to the wiring **811**, and the contact **812** is connected to a circuit for impressing a voltage which is not illustrated.

(397) When a voltage at a predetermined voltage value is impressed on the contact **812**, the voltage at the predetermined voltage value is impressed also on the wiring **811**, the light-shielding wall **751-1** and the light-shielding wall **751-2** which are connected to the contact **812**.

(398) Here, for example, let the voltage value of a voltage impressed on an anode **105** of an APD **21** be a voltage value A, and let the voltage value of a voltage impressed on a p-type semiconductor region **322** be a voltage value B, then the voltage value C of a voltage impressed on the light-shielding walls **751** is set in a relation of voltage value A > voltage value C > voltage value B.

(399) With such a voltage impressed, a state in which voltage is not abruptly varied but is varied gradually can be produced in the peripheral region **701e**.

(400) Note that, while a configuration in which the same voltage is impressed on the light-shielding wall **751-1** and the light-shielding wall **751-2** has been taken as an example in describing the example depicted in FIG. **59**, a configuration can be adopted in which voltages at different voltage values are individually impressed.

(401) For example, let the voltage value of a voltage impressed on the light-shielding wall **751-1** be a voltage value C-1, and let the voltage value of a voltage impressed on the light-shielding wall **751-2** be a voltage value C-2, then voltages such that voltage value A > voltage value C-1 > voltage value C-2 > voltage value B may be individually impressed on the light-shielding wall **751-1** and the light-shielding wall **751-2**.

(402) With such a configuration, it is possible to prevent (or alternatively, mitigate) a situation in which a high voltage impressed on the pixel region **A1** might generate a high-electric-field region in the isolation region, bringing about breakdown, and to secure an isolating property between pixels and the peripheral region and reliability thereof, against application of a high voltage.

Sixth Embodiment of Peripheral Region

(403) A peripheral region **701** in a sixth embodiment of the peripheral region will be described referring to FIG. **60**.

(404) A peripheral region **701f** depicted in FIG. **60** differs from the peripheral regions **701a'** to **701e** in the first to fifth embodiments of the peripheral region described above in that trenches **323** of different sizes are formed.

(405) Two trenches, namely, a trench **323-1'** and a trench **323-2'** are formed in the peripheral region **701f** in the sixth embodiment of the peripheral region illustrated in FIG. **60**. The trench **323-1'** is formed to be smaller in line width than the trench **323-2'**.

(406) Note that a case in which the trench **323-1'** is smaller than the trench **323-2'** in line width is depicted in FIG. **60**, but the trench **323-1'** may be formed to be greater than the trench **323-2'** in line width.

(407) In addition, though not illustrated here, in the case in which three or more trenches **323** are formed, the trenches **323** may be formed to be different in line width.

(408) FIG. **61** depicts a figure in which the part of the trench **323-2'** is depicted in an enlarged form. In the case in which the line width of the trench **323-2'** is increased, the inside of the trench **323-2'** may not be completely filled up with an insulating film and/or a light-shielding wall **751-2'** formed in the trench **323-2'**.

(409) In the case in which the light-shielding wall **751-2'** is formed in the trench **323-2'**, as exemplified in FIG. **61**, the inside of the trench **323-2'** may not be completely filled up with the material of the light-shielding wall **751-2'**, and a gap may be formed. Even in the case in which such a gap is formed, a configuration can be obtained in which an insulating film **831** is formed to form a lid on the gap.

(410) For instance, after the trench **323-2'** (light-shielding wall **751-2'**) is formed, the insulating film **831** can be formed by plasma CVD, sputtering or the like. As depicted in FIG. **60**, the insulating film **831** is formed also on a light-shielding film **402** and on an insulating wall **751-1'** (trench **323-1'**).

(411) The formation of the insulating film **831** in this way ensures that even if a gap is formed in a part, a configuration in which the part is covered with the insulating film **831** can be obtained. Therefore, it is possible to realize a product (in this case, a sensor chip **310**) which is resistant to variability of products and is high in yield.

(412) Note that at the time of production, a stopper **851** is formed at a lower portion of the trench **323-2'**, as depicted in FIG. **61**. The stopper **851** includes SiN, for example. At the time of forming the trench **323**, the stopper **851** is formed at a timing before boring, and then boring down to the stopper **851** is conducted. After the boring, the insulating film and/or the light-shielding wall is formed in the trench thus formed.

(413) While the stopper **851** has not been illustrated in the aforementioned embodiments, it is formed as a layer between the wiring layer **311** and the trench **323** also in, for example, the sensor chip **310** depicted in FIG. **60**. Note that a configuration in which the stopper **851** is not left in the sensor chip **310** can also be obtained, by such a treatment as polishing.

(414) In the case in which the stopper **851** is formed on a surface opposite to the light incidence surface side (a surface on the wiring layer **311** side) as depicted in FIG. **61**, the boring is conducted from the light incidence surface side (the upper side in the figure), to form the trench **323**. In the case in which the trench **323** is formed by conducting the boring from the light incidence surface side, the shape of the trench **323** in detail is as illustrated in FIG. **62**.

(415) Like FIG. **61**, FIG. **62** is a figure in which the part of the trench **323-2'** is depicted in an enlarged form. The trench **323-2'** depicted in FIG. **62** has a line width greater on the light incidence surface side (upper portion) than on the wiring layer **311** side (lower portion). Thus, the line width of the trench **323** is greater on the boring starting side and smaller on the boring ending side.

(416) When observing the trench **323** formed in a sensor chip **310** and comparing the line width at an upper portion of the trench **323** and the line width at a lower portion of the trench **323**, the side of a larger line width is the boring starting side, and the side of a smaller line width is the boring ending side. This can be made to be a criterion in determining the boring starting side at the time of production of a sensor chip **310**, based on the trench of the finished sensor chip **310**.

(417) With the trench **323** formed by conducting boring from the light incidence side, controllability of film thickness of a Si film at the time of production can be enhanced, and variability of pixel characteristics such as PDE and jitter can be suppressed.

Seventh Embodiment of Peripheral Region

(418) A peripheral region **701** in a seventh embodiment of the peripheral region will be described referring to FIG. **63**.

(419) A peripheral region **701g** depicted in FIG. **63** is different compared to the first to sixth embodiments of the peripheral region in that the trench is formed by conducting boring in a downward direction (from an upper side) and in an upward direction (from a lower side) in regard of the substrate. The first to sixth embodiments of the peripheral region have been described without any particular limitation on the boring direction at the time of producing the trench **323**.

(420) In the first to sixth embodiments of the peripheral region, the trench may be formed by conducting boring from the light incidence surface side, or the trench may be formed by conducting boring from the side of a surface connected to the wiring layer **311**.

(421) In the peripheral region **701g** depicted in FIG. **63**, trenches **872-1** and **872-2** are formed by performing boring from the light incidence surface side, whereas trenches **873-1** and **873-2** are formed by conducting boring from the surface on the side opposite to the light incidence surface. Thus, penetrating trenches can also be formed by conducting boring in the downward direction (from the upper side) and in the upward direction (from the lower side).

(422) With the penetrating trenches thus formed by conducting boring in the downward direction (from the upper side) and in the upward direction (from the lower side), the penetrating trenches can be formed even in the case in which, for example, the sensor chip **310** (exclusive of the wiring layer **311**) has a large-thickness structure.

(423) In addition, in the case in which the trenches are formed by conducting boring in different directions, the trenches can be filled up with different materials, to form trenches with different properties. For instance, in the example illustrated in FIG. **63**, a light-shielding wall **871-1** (**871-2**) is formed in the trench **872-1** (**872-2**), whereas the trench **873-1** (**873-2**) is filled up with an insulating material.

(424) Note that the trenches **872** and the trenches **873** can be filled up, or the light-shielding wall can be formed therein, by using the same material.

(425) By any of the first to seventh embodiments of the peripheral region, an isolating property between pixels and the peripheral region and reliability thereof can be secured, against application of a high voltage.

(426) In addition, any of the first to seventh embodiments of the peripheral region and any of the first to eighth embodiments of the isolation region can be combined with each other.

(427) For instance, in the case of a combination with the first embodiment of the isolation region depicted in FIG. **37**, the trench **412** is formed also in the isolation region **108a**. An influence of a dark current or white defect or the like arising from this trench **412** may be generated. The influence can be reduced by applying any of the first to seventh embodiments of the peripheral region.

(428) In other words, by applying any of the first to seventh embodiments of the peripheral region, an insulating property between pixels and the peripheral region and reliability thereof can be secured, against application of a high voltage, and the influence of the trench provided in the isolation region can also be reduced, since the insulating property between the pixels and the peripheral region and reliability thereof are secured.

Application to Imaging Apparatus

(429) The aforementioned APD **21** can be applied to a distance measuring device. Here, taking as an example a case in which the APD **21** is applied to a distance measuring device for measuring distance, an example of application of the APD **21** will be described.

(430) FIG. **64** is a figure illustrating a configuration of an embodiment of a distance measuring device to which the APD **21** according to the application of the present technology is applied. A distance measuring device **1000** depicted in FIG. **64** includes a light pulse transmitter **1021**, a light pulse receiver **1022**, and an RS flip-flop **1023**.

(431) A case of using a TOF (Time Of Flight) system as a method for measuring distance is taken as an example in the following description. The aforementioned APD **21** can be used as a TOF type sensor.

(432) A TOF type sensor is a sensor for measuring the distance from itself to an object by measuring the time taken until light emitted from itself impinges on the object, is reflected on the object and returns to itself. The TOF type sensor operates, for example, at timings depicted in FIG. **65**.

(433) Referring to FIG. **65**, an operation of the distance measuring device **1000** will be described. The light pulse transmitter **1021** emits light (light transmission pulse), based on a trigger pulse supplied thereto. The light emitted impinges on an object, and is reflected as reflected light, which is received by the light pulse receiver **1022**. The above-mentioned APD **21** can be used as the light

pulse receiver **1022**.

(434) The difference between the time at which the transmitted light pulse is emitted and the time at which the received light pulse is received corresponds to the distance between the distance measuring device **1000** and the object, or the time of flight TOF.

(435) The trigger pulse is supplied to the light pulse transmitter **1021**, and is supplied also to the flip-flop **1023**. With the trigger pulse supplied to the light pulse transmitter **1021**, a short-time light pulse is transmitted and is supplied to the flip-flop **1023**, whereby the flip-flop **1023** is reset.

(436) In the case in which the APD **21** is used as the light pulse receiver **1022**, a photon is generated when the received light pulse is received by the APD **21**. By the photon (electrical pulse) thus generated, the flip-flop **1023** is reset.

(437) By such an operation, a gate signal having a pulse width corresponding to the time of flight TOF can be produced. The gate signal thus produced is counted by use of a clock signal or the like, whereby the TOF can be calculated (outputted as a digital signal).

(438) In the distance measuring device **1000**, distance information is produced by the above-mentioned processing. The aforementioned APD **21** can be used for such a distance measuring device **1000**.

Example of Application to Mobile Body

(439) The technology according to the present disclosure (present technology) is applicable to various products. For example, the technology according to the present disclosure may be realized as a device to be mounted on a mobile body of any of such kinds as automobiles, electric vehicles, hybrid electric vehicles, motorcycles, bicycles, personal mobilities, airplanes, drones, ships, and robots.

(440) FIG. **66** is a block diagram depicting a general configuration example of a vehicle control system, which is an example of mobile body control system to which the technology according to the present disclosure can be applied.

(441) A vehicle control system **12000** includes a plurality of electronic control units interconnected through a communication network **12001**. In the example depicted in FIG. **66**, the vehicle control system **12000** includes a driving system control unit **12010**, a body system control unit **12020**, an exterior information detection unit **12030**, an interior information detection unit **12040**, and an integrated control unit **12050**. In addition, as functional components of the integrated control unit **12050**, there are depicted a microcomputer **12051**, an audio and image output section **12052**, and an in-vehicle network I/F (interface) **12053**.

(442) The driving system control unit **12010** controls operations of devices concerned to a driving system of the vehicle, according to various programs. For instance, the driving system control unit **12010** functions as a controller for a driving force generating device that generates a driving force of the vehicle, such as an internal combustion engine or a driving motor, a driving force transmission mechanism for transmitting the driving force to a wheel or wheels, a steering mechanism for regulating a steering angle of the vehicle, and a brake device for generating a braking force of the vehicle.

(443) The body system control unit **12020** controls operations of each of devices provided on a vehicle body, according to various programs. For example, the body system control unit **12020** functions as a controller for a keyless entry system, a smart key system, a power window device, or various lamps such as headlamp, tail lamp, brake lamp, turn signal lamps or fog lamp. In this case, electric waves emitted from a mobile device representing a key or signals from various switches may be inputted to the body system control unit **12020**. The body system control unit **12020** accepts these inputs of electric waves or signals, and controls a door lock device, the power window device, the lamps and the like of the vehicle.

(444) The exterior information detection unit **12030** detects information in the exterior of the vehicle on which the vehicle control system **12000** is mounted. For instance, an imaging section **12031** is connected to the exterior information detection unit **12030**. The exterior information

detection unit **12030** causes the imaging section **12031** to pick up an image of the vehicle exterior, and receives the picked-up image. The exterior information detection unit **12030** may perform object detection processing or distance measurement processing in regard of humans, vehicles, obstacles, signs, characters on the road surface, etc., based on the received image.

(445) The imaging section **12031** is an optical sensor that receives light and outputs an electrical signal according to the amount of light received. The imaging section **12031** can output the electrical signal as an image, or can output the electrical signal as information regarding distance measurement. In addition, the light received by the imaging section **12031** may be visible light, or invisible light such as infrared rays.

(446) The interior information detection unit **12040** detects information in the vehicle interior. The interior information detection unit **12040** is connected, for example, with a driver condition detection section **12041** for detecting the condition of a driver. The driver condition detection section **12041** includes, for example, a camera for imaging the driver, and the interior information detection unit **12040** may calculate the degree of fatigue or the degree of concentration of the driver, and may determine whether or not the driver is dozing in a sitting position based on the detected information inputted from the driver condition detection section **12041**.

(447) The microcomputer **12051** can calculate control setpoints for the driving force generating device, the steering mechanism or the brake device, based on exterior or interior information acquired by the exterior information detection unit **12030** or the interior information detection unit **12040**, and can output control commands to the driving system control unit **12010**. For instance, the microcomputer **12051** can perform cooperative control for the purpose of realizing the function of ADAS (Advanced Driver Assistance System) that includes vehicle collision avoidance or shock mitigation, following traveling based on inter-vehicle distance, vehicle speed maintaining running, vehicle collision alarm, lane departure alarm, etc.

(448) In addition, the microcomputer **12051** controls the driving force generating device, the steering mechanism, the brake device or the like, based on the information regarding the surroundings of the vehicle acquired by the exterior information detection unit **12030** or the interior information detection unit **12040**, whereby cooperative control aimed at automatic driving for autonomously traveling without depending on the driver's operation, or the like can be performed.

(449) Besides, the microcomputer **12051** can output control commands to the body system control unit **12020**, based on the exterior information acquired by the exterior information detection unit **12030**. For instance, the microcomputer **12051** can perform cooperative control aimed at anti-glare such as switching from a high beam mode to a low beam mode, by controlling the headlamp according to the position of the leading vehicle or the oncoming vehicle that is detected by the exterior information detection unit **12030**.

(450) The audio and image output section **12052** emits at least one output signal of audio and image signals to an output device capable of giving audio or visual information to the rider on the vehicle or to the exterior of the vehicle. In the example depicted in FIG. **66**, an audio speaker **12061**, a display section **12062** and an instrument panel **12063** are depicted as examples of the output device. The display section **12062** may include at least one of an on-board display and a head-up display, for example.

(451) FIG. **67** depicts an example of layout position of the imaging section **12031**.

(452) In FIG. **67**, imaging sections **12101**, **12102**, **12103**, **12104**, and **12105** are provided as the imaging section **12031**.

(453) The imaging sections **12101**, **12102**, **12103**, **12104**, and **12105** are provided, for example, at such positions as a front nose, side mirrors, a rear bumper, or a back door of the vehicle **12100**, and an upper portion of a front glass in the vehicle compartment. The imaging section **12101** provided at the front nose and the imaging section **12105** provided at the upper portion of the front glass in the vehicle compartment acquire mainly an image of the front side of the vehicle **12100**. The imaging sections **12102** and **12103** provided on the side mirrors acquire mainly images on the

lateral sides of the vehicle **12100**. The imaging section **12104** provided on the rear bumper or the back door acquires mainly an image of the rear side of the vehicle **12100**. The imaging section **12105** provided at the upper portion of the front glass in the vehicle compartment is used primarily for detection of a leading vehicle, pedestrians, obstacles, signals, traffic signs, traffic lanes, etc. (454) Note that FIG. **67** depicts an example of imaging ranges of the imaging sections **12101** to **12104**. An imaging range **12111** represents an imaging range of the imaging section **12101** provided at the front nose; imaging ranges **12112** and **12113** represent imaging ranges of the imaging sections **12102** and **12103** provided individually on the side mirrors; and an imaging range **12114** represents an imaging range of the imaging section **12104** provided on the rear bumper or the back door. For instance, where the image data picked up by the imaging sections **12101** to **12104** are superimposed, a bird's-eye view image as if obtained by viewing the vehicle **12100** from above is obtained.

(455) At least one of the imaging sections **12101** to **12104** may have a function to acquire distance information. For instance, at least one of the imaging sections **12101** to **12104** may be a stereo camera including a plurality of imaging elements, or may be an imaging element having pixels for phase difference detection.

(456) For example, the microcomputer **12051** may obtain the distance to each solid object within the imaging ranges **12111** to **12114** and time variation of this distance (relative velocity with reference to the vehicle **12100**), based on the distance information acquired from the imaging sections **12101** to **12104**. By this, particularly, a solid object which is on the traveling path of the vehicle **12100**, is the nearest to the vehicle **12100** and is running at a predetermined speed (for example, not less than 0 km/h) in substantially the same direction as the vehicle **12100** can be extracted as the leading vehicle. Further, the microcomputer **12051** can set an inter-vehicle distance to be preliminarily secured between the vehicle **12100** and the leading vehicle, and can perform an automatic brake control (inclusive of a following stop control) or an automatic acceleration control (inclusive of a following start control). In this way, it is possible to perform cooperative control aimed at automatic driving for autonomously traveling without depending on the driver's operation.

(457) For instance, the microcomputer **12051** can extract solid object data concerning solid objects while classifying them into two-wheel vehicles, ordinary-sized vehicles, heavy duty vehicles, pedestrians, electric poles, and other solid objects, based on the distance information obtained from the imaging sections **12101** to **12104**, and can use the extracted solid object data for automatic avoidance of obstacles. For example, the microcomputer **12051** discriminates the obstacles in the surroundings of the vehicle **12100** into obstacles that can be visually recognized by the driver of the vehicle **12100** and obstacles that are difficult to visually recognize. Then, the microcomputer **12051** determines collision risk representing the degree of danger of collision with each obstacle. In a situation in which the collision risk is not less than a preset value and there is a possibility of collision, the microcomputer **12051** causes an alarm to be outputted to the driver through the audio speaker **12061** and/or the display section **12062**, or causes forced deceleration or avoiding steering to be performed through the driving system control unit **12010**, whereby driving assistance for avoiding collision can be performed.

(458) At least one of the imaging sections **12101** to **12104** may be an infrared camera for detection of infrared rays. For example, the microcomputer **12051** can recognize a pedestrian by determining whether or not a pedestrian is present in the images picked up by the imaging sections **12101** to **12104**. Such recognition of a pedestrian is conducted, for example, by a procedure of extracting characteristic points in the images picked up by the imaging sections **12101** to **12104** as infrared cameras, and a procedure of determining whether or not an object is a pedestrian by performing pattern matching processing on a series of characteristic points representing the profile of the object. When the microcomputer **12051** determines that a pedestrian is present in the images picked up by the imaging sections **12101** to **12104** and recognizes the pedestrian, the audio and image output section **12052** controls the display section **12062** such that a rectangular profile line for

emphasis is superimposed on the recognized pedestrian. In addition, the audio and image output section **12052** may control the display section **12062** such that an icon or the like representing a pedestrian is displayed at a desired position.

(459) Note that the effects described herein are merely exemplary and are not restrictive, and other effects may exist.

(460) Note that embodiments of the present technology are not restricted to the above-described embodiments, and various modifications are possible without departing from the scope of the gist of the present technology.

(461) Note that the present technology can take the following configurations.

(462) (1)

(463) A sensor, comprising: a first substrate including at least a first pixel, the first pixel including: an avalanche photodiode to convert incident light into electric charge and including an anode and a cathode, wherein the cathode is in a well region of the first substrate; an isolation region that isolates the well region from at least a second pixel that is adjacent to the first pixel; and a hole accumulation region between the isolation region and the well region, wherein the hole accumulation region is electrically connected to the anode.

(464) (2)

(465) The sensor of (1), further comprising: a lens on the first substrate to focus incident light toward the well region, wherein the hole accumulation region includes a portion between the lens and the well region.

(466) (3)

(467) The sensor of (2), further comprising: a second substrate including: a first contact electrically connected to the anode; a second contact electrically connected to the cathode; a first wiring layer electrically connected to the first contact; and a second wiring layer electrically connected to the second contact.

(468) (4)

(469) The sensor of (3), further comprising: a third substrate on the second substrate and electrically connected to the first wiring layer and the second wiring layer.

(470) (5)

(471) The sensor of (1), wherein the well region includes a first region of a first conductivity type, a second region of a second conductivity type and in contact with the first region, and a third region of the second conductivity type or the first conductivity type in contact with the hole accumulation region.

(472) (6)

(473) The sensor of (5), wherein if the third region is of the second conductivity type, the second region and the third region have different impurity concentrations of the second conductivity type, and wherein, if the third region is of the first conductivity type, the first region and the third region have different impurity concentrations of the first conductivity type.

(474) (7)

(475) The sensor of (6), wherein the first conductivity type is n-type and the second conductivity type is p-type.

(476) (8)

(477) The sensor of (6), wherein, in a plan view, the first region is surrounded by the third region, and the third region is surrounded by the anode, and the anode is surrounded by the isolation region.

(478) (9)

(479) The sensor of (2), further comprising: an oxide film between the isolation region and the hole accumulation region, and between the hole accumulation region and the lens.

(480) (10)

(481) The sensor of (2), further comprising: a conductive film between the isolation region and at

least the second pixel.

(482) (11)

(483) The sensor of (2), further comprising: a light shielding film on a same surface of the first substrate as the lens; a first contact electrically connected to the light shielding film and configured to receive a bias voltage; and a second contact electrically connected to the isolation region and configured to receive the bias voltage.

(484) (12)

(485) A sensor, comprising: a substrate including at least one pixel, the at least one pixel including: an avalanche photodiode to convert incident light into electric charge and including a cathode of a first conductivity type and an anode of a second conductivity type, wherein the cathode is in a well region of the first conductivity type or the second conductivity type, and wherein the anode is at a perimeter of the well region; a first material of the second conductivity type around the well region, wherein the first material is electrically connected to the anode; and an insulating material around the first material and the anode.

(486) (13)

(487) The sensor of (12), further comprising: a lens on the substrate, wherein the first material includes a portion between the lens and the well region.

(488) (14)

(489) The sensor of (13), further comprising: an oxide film between the insulating material and the first material, and between the first material and the lens.

(490) (15)

(491) The sensor of (12), further comprising: a metal film around the insulating material.

(492) (16)

(493) The sensor of (12), wherein the well region includes a first area of the second conductivity type and in contact with the cathode, and a second area of the first conductivity type or the second conductivity type in contact with the cathode and the anode.

(494) (17)

(495) The sensor of (14), wherein, the second area is the second conductivity type, and the first area and the second area have different impurity concentrations of the second conductivity type.

(496) (18)

(497) A sensor, comprising: a first substrate including at least a first pixel in a pixel region, the first pixel including: an avalanche photodiode to convert incident light to electric charge and including a first region, a second region, and a third region, wherein the first region is a first conductivity type, the second region is a second conductivity type, the third region is the first conductivity type or the second conductivity type, and the first and third regions are stacked; an isolation region that isolates the third region from at least a second pixel, wherein the isolation region penetrates through the first substrate; and a charge accumulation region of the second conductivity type between the third region and the isolation region, wherein the charge accumulation region is electrically connected to the second region.

(498) (19)

(499) The sensor of (18), wherein the charge accumulation region is between the isolation region and the third region and extends from a first surface of the first substrate toward a second surface of the first substrate opposite the first surface, wherein the first surface is at a light incident side of the first substrate.

(500) (20)

(501) The sensor of (18), wherein the isolation region includes: a metal film penetrated through the first substrate.

(502) (21)

(503) The sensor of (18), further comprising: a fourth region between the first region and third region, wherein the fourth region is the second conductivity type.

(504) (22)

(505) The sensor of (18), further comprising: a lens on the first substrate to focus incident light toward the third region.

(506) (23)

(507) The sensor of (18), further comprising: a second substrate including: a first contact electrically connected to the first region; a second contact electrically connected to the second region; a first wiring electrically connected to the first contact; and a second wiring electrically connected to the second contact.

(508) (24)

(509) The sensor of (23), further comprising: a third substrate on the second substrate and electrically connected to the first wiring and the second wiring.

(510) (25)

(511) The sensor of (24), wherein the third substrate includes a third wiring and a fourth wiring, the first wiring is electrically connected to the third wiring, and the second wiring is electrically connected to the fourth wiring.

(512) (26)

(513) The sensor of (18), further comprising: an oxide film between the isolation region and the charge accumulation region.

(514) (27)

(515) The sensor of (18), wherein if the third region is the first conductivity type, the first region and the third region have different impurity concentrations of the first conductivity type, wherein if the third region is the second conductivity type, the second region and the third region have different impurity concentrations of the second conductivity type, and wherein the second region and the charge accumulation region have different impurity concentrations of the second conductivity type.

(516) (28)

(517) The sensor of (18), further comprising: a lens on the first substrate to focus incident light toward the third region; a light shielding film on a same surface of the first substrate as the lens; a first contact electrically connected to the light shielding film and configured to receive a bias voltage; and a second contact electrically connected to the isolation region and configured to receive the bias voltage.

(518) (29)

(519) The sensor of (24), further comprising: a trench region in the first substrate, the trench region being in a peripheral region of the first substrate other than the pixel region.

(520) (30)

(521) The sensor of (29), further comprising: a connection via penetrating through the first and second substrates to a wiring of the third substrate, wherein the trench region is between the pixel region and the connection via.

(522) (31)

(523) A photodetector comprising: a high-electric-field region; an isolation region for isolation from neighboring pixels; and a hole accumulation region that traps electrons on a side wall of the isolation region, wherein the hole accumulation region is electrically connected with an anode.

(524) (32)

(525) The photodetector according to (31), wherein the hole accumulation region is formed also on a back surface side.

(526) (33)

(527) The photodetector according to (31), wherein the hole accumulation region is a p-type semiconductor region.

(528) (34)

(529) The photodetector according to (31), wherein the hole accumulation region is formed by

induction of holes by a negative fixed charge film.

(530) (35)

(531) The photodetector according to (31), wherein the hole accumulation region is formed by induction of holes by impressing of a voltage on a metallic film formed in the isolation region.

(532) (36)

(533) The photodetector according to (31), wherein of a cathode constituting the high-electric-field region, other portion than a portion connected with a contact is embedded in a substrate, a hole accumulation region is formed at a surface of the substrate in a region other than the portion connected with the contact, and a potential of the hole accumulation region formed at the surface of the substrate is different from a potential of the anode.

(534) (37)

(535) The photodetector according to (36), wherein the potential of the hole accumulation region formed at the surface of the substrate is identical to a ground potential or a potential of the cathode.

(536) (38)

(537) The photodetector according to (36), wherein the portion connected with the contact is formed in a substantially central portion of the high-electric-field region.

(538) (39)

(539) The photodetector according to (36), wherein the portion connected with the contact is formed near an end of the high-electric-field region, and the hole accumulation region formed at the surface of the substrate is formed in a central portion of the high-electric-field region.

(540) (40)

(541) The photodetector according to (31), wherein the anode is formed at a portion adjacent to the isolation region.

(542) (41)

(543) A photodetector comprising: a high-electric-field region; an isolation region for isolation from neighboring pixels; and an electron accumulation region that traps holes on a side wall of the isolation region, wherein the electron accumulation region is electrically connected with a cathode.

(544) (42)

(545) The photodetector according to (41), wherein the electron accumulation region is formed also on a back surface side.

(546) (43)

(547) The photodetector according to (41), wherein the electron accumulation region is an n-type semiconductor region.

(548) (44)

(549) The photodetector according to (41), wherein the electron accumulation region is formed by induction of electrons by a positive fixed charge film.

(550) (45)

(551) The photodetector according to (41), wherein the electron accumulation region is formed by induction of electrons by impressing of a voltage on a metallic film formed in the isolation region.

(552) (46)

(553) The photodetector according to (41), wherein of an anode constituting the high-electric-field region, other portion than a portion connected with a contact is embedded in a substrate, an electron accumulation region is formed at a surface of the substrate in a region other than the portion connected with the contact, and a potential of the electron accumulation region formed at the surface of the substrate is different from a potential of the cathode.

(554) (47)

(555) The photodetector according to (46), wherein the potential of the electron accumulation region formed at the surface of the substrate is identical to a ground voltage or a voltage of the anode.

(556) (48)

(557) The photodetector according to (46), wherein the portion connected with the contact is formed in a substantially central portion of the high-electric-field region.

(558) (49)

(559) The photodetector according to (46), wherein the portion connected with the contact is formed near an end of the high-electric-field region, and the electron accumulation region formed at the surface of the substrate is formed in a central portion of the high-electric-field region.

(560) (50)

(561) The photodetector according to (41), wherein the cathode is formed at a portion adjacent to the isolation region.

REFERENCE SIGNS LIST

(562) **21** APD **23** On-chip lens **101** n-type semiconductor region **102** p-type semiconductor region **103** Well layer **104** Contact **105** Anode **106** Contact **107** Hole accumulation region **108** Isolation region **151** Fixed charge film **171** Metallic film

Claims

1. A light detecting device, comprising: a first semiconductor layer, including: a first avalanche photodiode, including: a first semiconductor region of a first conductivity type; and a second semiconductor region of a second conductivity type, wherein the first avalanche photodiode is disposed in a pixel region of the light detecting device; a first isolation region; a third semiconductor region of the second conductivity type, wherein the third semiconductor region is disposed in a peripheral region; and a light shielding film, wherein the first isolation region is disposed between the pixel region and the peripheral region, wherein a first portion of the light shielding film is disposed in the first isolation region and a second portion of the light shielding film is disposed on the third semiconductor region, and wherein a portion of the light shielding film is in direct contact with at least a portion of a surface of the third semiconductor region.
2. The light detecting device of claim 1, further comprising: a wiring layer, the wiring layer including: a first wiring, wherein the first wiring is coupled to the first semiconductor region; a second wiring, wherein the second wiring is coupled to the second semiconductor region; and a third wiring, wherein the third wiring is coupled to the third semiconductor region.
3. The light detecting device of claim 2, wherein a first voltage applied to the second wiring is same as a second voltage applied to the third wiring.
4. The light detecting device of claim 2, wherein the wiring layer is on a side of the first semiconductor layer opposite a light incident surface side of the first semiconductor layer.
5. The light detecting device of claim 1, wherein the light shielding film includes at least one of tungsten, aluminum, copper, or polysilicon.
6. The light detecting device of claim 1, wherein the first conductivity type is opposite the second conductivity type.
7. The light detecting device of claim 2, wherein the first semiconductor layer is part of a first chip, the light detecting device further comprising: a second chip.
8. The light detecting device of claim 7, wherein the second chip includes a fourth wiring, and a fifth wiring, wherein the fourth wiring is electrically connected to the first wiring, and wherein the fifth wiring is electrically connected to the second wiring.
9. The light detecting device of claim 8, wherein the second chip is stacked on the first chip.
10. The light detecting device of claim 8, wherein the first, second, third, fourth, and fifth wirings are one of copper or aluminum.
11. The light detecting device of claim 8, wherein the first wiring is directly bonded to the fourth wiring, and wherein the second wiring is directly bonded to the fifth wiring.
12. The light detecting device of claim 2, wherein the first, second, and third wirings are in a same layer of the wiring layer.

13. The light detecting device of claim 1, wherein the first isolation region extends from a light incident surface of the first semiconductor layer to a surface of the first semiconductor layer opposite the light incident surface.
 14. The light detecting device of claim 1, wherein the light shielding film is electrically conductive.
 15. The light detecting device of claim 14, wherein the light shielding film is formed from a metal.
 16. The light detecting device of claim 14, wherein the light shielding film electrically connects the first isolation region and the third semiconductor region.
 17. The light detecting device of claim 1, wherein the first isolation region includes a trench filled with an electrically conductive material.
 18. The light detecting device of claim 1, wherein the first isolation region surrounds the first semiconductor region.
 19. The light detecting device of claim 18, further comprising: an on-chip lens, wherein the on-chip lens is disposed on a light incident surface side of the first semiconductor layer, and wherein the on-chip lens is adjacent an area encompassing the first avalanche photodiode.
 20. The light detecting device of claim 1, wherein a third portion of the light shielding film is disposed on a planarizing film.
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